



## D3.3

### Report on the outcomes of hazard and negative impacts



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## Abbreviations

List of abbreviations and acronyms featured throughout the document:

**CAES:** Spanish Energy Saving Certificates

**CAP:** Common Agricultural Policy

**DGs:** Disadvantaged Groups

**DE:** Germany

**DoA:** Description of Action

**ES:** Spain

**ETS:** Emission Trading System

**ETUI:** European Trade Union Institute

**EU:** European Union

**EUT:** Eurecat

**EV:** Electric Vehicles

**HU:** Hungary

**IE:** Ireland

**IT:** Italy

**IZ:** Inclusionary Zoning

**KPIs:** Key Performance Indicators

**MaaS:** Mobility-as-a-service

**MLP:** Multi-Level Perspective

**NA:** North Africa

**OECD:** Organisation for Economic Co-operation and Development

**PESTLE:** Political, Economic, Social, Technological, Legal and Environmental

**PT:** Portugal

**RE:** Renewable Energy

**SDGs:** Sustainable Development Goals (SDGs).

**TT:** Twin Transition

## Summary

The deliverable D3.3 reports on the outcomes of hazards and negative impacts associated with the twin transition in Europe, with a particular focus on how these processes affect Disadvantaged Groups across four critical sectors: energy, housing and built environment, mobility and transport, and agriculture and food. Building on prior work in D3.1 and D3.2, this report maps hazards and links them to negative impacts, prioritizing them according to their likelihood and severity.

The analysis demonstrates that hazards rarely occur in isolation. Instead, they interact and reinforce one another, producing cumulative disadvantages that deepen existing inequalities. Vulnerable groups most exposed to these risks include low-income households, migrants and refugees, older workers, rural residents, small-scale farmers, people with disabilities, and digitally excluded individuals. The findings show that the twin transition will only succeed if social justice and fairness are placed at its core.

The methodology combines three complementary approaches: a literature review of risks linked to the twin transition; 34 semi-structured interviews with potentially vulnerable and disadvantaged groups across five Member States; and a Delphi survey with experts to rank hazards according to probability and impact. This evidence triangulation is supported by work developed in D3.1 and D3.2 to ensure robustness.

The findings reveal that while the twin transition promises environmental and technological progress, it also risks reproducing or exacerbating inequalities if not proactively governed. Robust, equity-centred measures are necessary to avoid disproportionate costs in terms of job loss, energy and housing insecurity, financial strain, and digital exclusion.

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## 1. Introduction

### 1.1 Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., the FITTER ecosystem) that pivotally includes a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System. Through the appliance of a co-creation methodological approach, the ecosystem aims to enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies for the assessment of inequalities and social exclusion risks. The Digital Platform within the ecosystem aims to provide proposed mitigation measures and better practice guides that can be used to curve potential negative effects of policies on identified at-risk groups. In addition, the FITTER ecosystem incorporates wide platform connectivity capability, which helps to support the needs of the engaging stakeholders within the community of the FITTER Cluster Group (CG). The group includes key national and pan-European members such as policymakers and civil society organizations.

### 1.2 Purpose and scope of the Deliverable

This deliverable presents the outcomes of Task 3.3, which focuses on mapping hazards and negative impacts of the twin transition across the three future scenarios developed under T3.2 and documented in D3.2 (optimistic, neutral, pessimistic). The analysis examines these hazards in relation to three vulnerability dimensions (finance, employment and skills) and assesses their potential impacts on the Disadvantaged Groups (DGs) identified in D3.2 and further expanded in this report. The four sectors under analysis are: energy, mobility and transport, housing and built environment, and agriculture and food.

For the purposes of this deliverable, the emphasis is placed on the neutral and pessimistic scenarios, in order to capture the most pressing risks for vulnerable groups.

The main objectives of D3.3 are to:

- Identify key hazards arising under each future scenario in relation to each DG.
- Link hazards to negative impacts, and negative impacts to Disadvantaged Groups
- Prioritise hazards based on their likelihood and potential severity of impact.
- Provide structured inputs for T4.2 FITTER Labs and WP4 intersectional mitigation plans.

### 1.3 Structure of the deliverable

The deliverable is structured as follows:

- Section 1 introduces the context, providing an overview of the FITTER-EU project and the purpose and scope of the deliverable
- Section 2 develops the theoretical concept used for the analysis
- Section 3 outlines the methodology applied to conduct Task 3.3 and the Deliverable.
- Section 4 presents a complete literature review used to feed the Deliverable
- Section 5 shows the results of semi-structured interviews with potential Disadvantaged Groups

- Section 6 shows the main results of the analysis, divided by sectors.
- Section 7 presents the findings of the analysis in a synthesized and prioritized way.
- Section 8 draws the main conclusions of the deliverable.
- Section 9 highlights the main limitations encountered during the research process.
- Section 10 provides the list of references in APA format.

## 2. Theoretical concepts

As described in the Description of Action (DoA), vulnerabilities in the context of the twin transition—across the three dimensions addressed by FITTER (labour market, financial resources, and skills)—are understood as low capabilities to adapt to change and achieve wellbeing. For example, a lack of digital skills constitutes a vulnerability when adapting to the demands of the twin transition.

**Hazards** are defined as intervening factors that mediate between vulnerability and negative impacts, exacerbating or widening existing inequalities, or creating new ones. For instance, the closure of physical bank branches may act as a hazard for individuals with low digital skills, limiting their access to essential services.

**Negative impacts** refer to the consequences of hazards on vulnerable individuals or population groups. For example, if people with low digital skills are unable to access certain financial products, the resulting exclusion could have a detrimental effect on their personal economies.

**Disadvantaged Groups (DGs)** are those who are exposed to vulnerabilities and hazards arising from the twin transition, with the potential to trigger significant and often intertwined negative impacts on their employment conditions, skills, or income levels.

A diagram reflecting the relationships between these concepts is included in Figure 1.

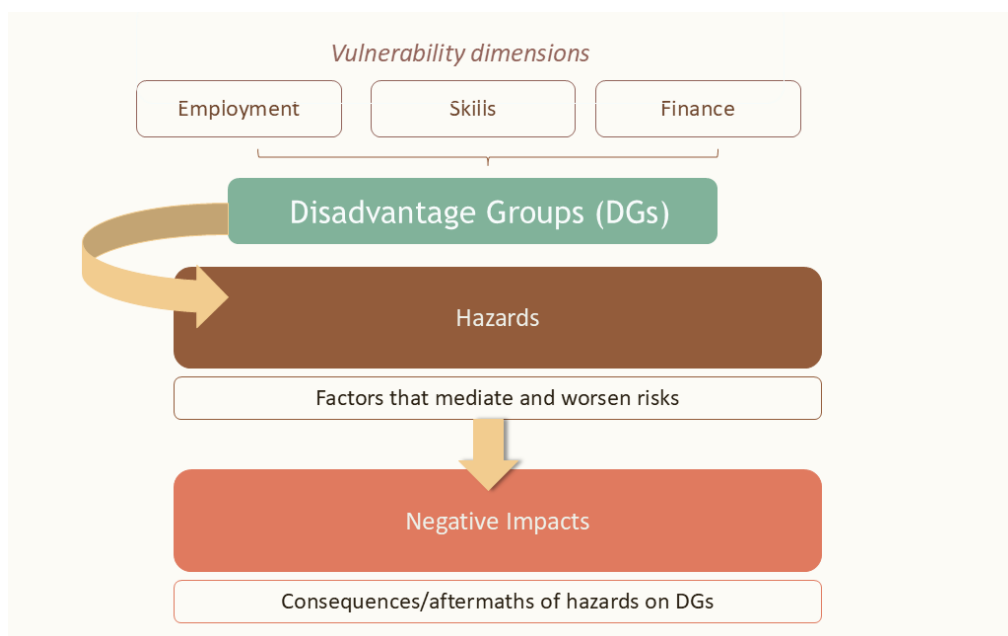


Figure 1: Theoretical concepts diagram



### 3. Methodology

The methodology for Task 3.3 follows a three-step process designed to identify Disadvantaged Groups (DGs), map hazards arising from the twin transition, and prioritise these hazards according to their potential negative impacts and likelihood. This process integrates multiple sources of evidence, ensuring that the analysis is grounded in both qualitative and quantitative insights.

#### Step 1: Identification of DGs and negative impacts

The first stage builds on the results of previous tasks (D3.1 and D3.2) and incorporates insights from three complementary sources:

- **Interview insights:** Semi-structured interviews with experts (included in D3.2) and potentially vulnerable individuals (included in this deliverable in section 5) capturing perceptions, lived experiences, and sector-specific challenges.
- **Delphi survey results:** Expert assessments gathered through iterative surveys, highlighting key trends, uncertainties, and risks under different future scenarios.
- **Literature review:** Review of academic and policy sources to validate and complement the empirical findings collected in D3.1

DGs and associated negative impacts are contextualised to the future scenarios developed under T3.2, ensuring that their vulnerabilities are analysed within the framework of sector-specific developments for the year 2050.

#### Step 2: Identification of hazards

Hazards are defined as intervening factors that mediate between vulnerability and negative impacts, potentially exacerbating existing inequalities or creating new ones. The identification process combines:

- **Literature review** of previous research on twin transition risks across environmental, social, and economic dimensions.
- **Identification of results** of D3.1 and D3.2, as previously mentioned.

#### Step 3: Prioritisation of hazards

The hazards identified are assessed and ranked based on the main criteria linked to the Delphi Survey developed in D3.2:

1. **Likelihood** of occurrence within the period and context of the scenarios.
2. **Potential Impact** on Disadvantaged Groups

This is done through the calculation of a risk index, being the formula:

$$\text{Risk index} = (\text{Average Likelihood consensus} \times \text{Average Impact consensus})^{12}$$

<sup>1</sup> The averages are introduced to make the different sections comparable, due to the difference in answers in each sector. Thus, maximum average achievable is 5, as there are 5 potential answers that will represent the score for each statement: Very high (5), high (4), neutral (3), low (2), very low (1). Therefore, the maximum risk index is 25, and the minimum is 1.

<sup>2</sup> If there is more than 1 Delphi survey statement related, the risk index will be calculated through the average of the risk indexes of both statements

Hazards are prioritized according to their risk index following a decreasing logic (hazards with a higher risk index will be prioritized first).

## 4. Literature review

The twin transition refers to the simultaneous pursuit of a green (environmental) and a digital transition. In Europe, this concept has become central to policy agendas, especially since the European Green Deal (targeting climate neutrality by 2050) and the Digital Strategy were launched around 2019-2020 (Pettinicchio, 2023; Aloisi, 2025a). The EU's post-COVID recovery plan explicitly ties recovery funds to advancing both green and digital objectives in parallel (Universitat Oberta de Catalunya, 2025). The rationale is that digital innovation can enable and accelerate sustainability efforts, while the green transition can shape the direction of technological innovation (Kirov, 2023).

However, as policies favouring these twin transitions are implemented, various hazards and negative consequences have been identified. The FITTER project defines a hazard as the intervening factors mediating between vulnerability and negative impacts, which exacerbate or widen existing inequalities, or create new ones. Policymakers and scholars increasingly warn that without careful management, the twin transition could produce new social inequalities and economic disruptions, as well as environmental problems, even as it solves others (Aloisi, 2025a; López, 2024). Indeed, the European Commission has stated that “The twin transitions will be fair or will not be”, highlighting that success depends on addressing these concerns (European Commission, 2022). This review explores the literature on the hazards linked to the twin transition in Europe. Understanding these risks is crucial to make sure that the green and digital transformations are effective and equitable.

One paradox of the twin transition is that digital technologies, which are intended to help solve environmental challenges, can themselves introduce significant environmental costs. Studies find that large-scale digitalization comes with “very high costs in terms of energy, water and rare materials” and can “lead to significant environmental problems” if expanded without restraint (Universitat Oberta de Catalunya, 2025). The ICT sector's footprint is substantial: by some estimates it accounts for 5-9% of global electricity use and around 3% of greenhouse gas emissions (Aloisi, 2025b). Data centres, cloud computing, blockchain and cryptocurrency networks, and AI training models consume enormous energy and water resources, often relying on non-renewable energy and intensive cooling systems (Muñoz de Bustillo, 2024). In addition, the mass deployment of digital devices generates mounting electronic waste (e-waste), and the extraction of rare minerals for digital hardware and electric batteries, (such as lithium, cobalt, and rare earth elements) causes pollution and habitat destruction (Muñoz de Bustillo, 2024). For instance, severe oil and water pollution from mining of raw materials is cited as a side effect of producing hardware for both renewable energy and digital infrastructure. Muñoz de Bustillo (2024, p.1) highlights “the high-energy consumption associated with digital technologies, the generation of e-waste, and the environmental impact of cryptocurrencies and artificial intelligence” as key negative consequences of digitalization for the environment.

Efficiency gains from digital solutions do not automatically translate into net environmental effects; digitalisation is not a panacea for sustainability (Muñoz de Bustillo, 2024, p.3). In some cases, digital solutions enable higher consumption levels that offset the savings. For example, smarter logistics and autonomous driving might reduce energy use per trip, but if they also make travel cheaper or more convenient, people may travel more, eroding the emission cuts (Muñoz de Bustillo, 2024). The JRC warns of “significant backlash due to the existence of rebound effects”, where improvements (like better fuel efficiency or automated energy management) lead to behavioural shifts (driving longer distances, leaving lights on longer, etc.) that undermine the initial purpose (Muñoz de Bustillo, 2024, p.1). Similarly, digital

platforms can increase consumerism and waste. For instance, e-commerce causes more packaging waste and product returns, or micromobility apps can increase traffic if they displace public transport usage (Muñoz de Bustillo, 2024). These unintended outcomes mean that without safeguards, digitalization can end up “nullifying net emission reductions” that green policies achieve (Bidmon, Piscicelli, & Sussha, 2024).

Another environmental hazard lies in the rush for critical raw materials required for green technologies (such as batteries, solar panels, or wind turbines) and digital devices. Europe’s green transition dramatically increases demand for materials like lithium, nickel, cobalt, or rare earths (Granvik et al, 2025). This has two consequences: first, it creates supply-chain vulnerabilities, and second, it externalizes environmental and social costs to mining regions, often in the Global South. Analysts caution that the “emphasis on supply risks overlooking conflicts that mining may give rise to, including land disputes, social unrest and displacement of local communities” (Bogojevic, 2024, p.600). For example, speeding up mining permits without proper safeguards “risks bypassing environmental and social safeguards reflects a narrow framing that fails to account for broader socio-political consequences” and can lead to deforestation, water pollution, and human rights abuses in communities where these minerals are extracted (European Environmental Bureau, 2023). Thus, an ill-managed green boom could simply ride one form of environmental degradation (fossil fuel extraction) for another (mining impacts), shifting burdens rather than eliminating them.

Finally, a subtler hazard is encountered: environmental problem-shifting due to digital solutionism. Kovacic et al. (2024, p.2273) argue that EU policy discourse tends to put “environmental governance at the service of (...) the new digital sectors (...) possibly redefining the scope of environmental governance away from environmental issues and towards promoting the sustainability of the new digital sectors”. This can mean that problems susceptible to digital fixes are prioritized over more urgent but less tech-solvable issues. For instance, climate change data management might get attention, while matters biodiversity loss or soil degradation receive less focus (Kovacic, 2024 as cited in López, 2024). The hazard here is that critical ecological challenges may be neglected because they don’t align neatly with digital transition goals, effectively putting some environmental problems on hold (Kovacic, 2024).

A major category of twin transition hazards is social justice and equity. Multiple sources stress that both the green and digital transitions risk exacerbating inequalities – geographical, gender-based, and sectoral (Aloisi, 2025a). Without deliberate intervention, the benefits of these transitions may be captured by certain groups or regions, while costs fall disproportionately on others.

Regional and spatial disparities are a prime concern in Europe. Prosperous, innovation-driven regions are better positioned to attract green tech investments and develop digital infrastructures, whereas less-developed or industrial regions may struggle. A 2023 study by the Vienna Institute for International Economic Studies found that “greening and digitalizing the economy will likely widen the gap between rich and poor regions in Europe” (Maucorps et al., 2023, p.1). Maucorps et al. (2023) also found that regions that are most ready for the twin transition, which are mainly metropolitan regions, are the most likely to benefit economically from the digital and green transition versus agricultural regions and regions specializing in low-tech manufacturing. Regions with higher incomes and advanced industries can more readily adopt digital solutions and shift to renewable energy, creating growth and jobs from the twin transition. In contrast, poorer regions (especially those reliant on legacy industries like coal, steel, or traditional manufacturing) face the loss of old jobs without immediately enjoying the new opportunities, losing cohesion relative to wealthier, more prepared areas (Bertelsmann Stiftung, 2022).

Related to regional gaps are urban-rural divides. Rural communities often have less access to digital connectivity and fewer green jobs alternatives. Many protests have been related to rural grievances, such as farmers and villagers who feel left behind by climate policies. The emblematic case is France’s Gilets

Jaunes movement. This movement began in 2018 against a fuel carbon tax that hit rural commuters hard, and it soon became “a powerful symbol of why social justice must always be at the heart of environmental justice” (Martin, 2019). Far from rejecting climate action, the protesters objected that the flat fuel tax was regressive, burdening lower-income, car-dependent people in rural areas while urban elites were less affected (“why should the costs of climate action fall mostly on the poor?”) (Martin, 2019). The Gilets Jaunes showed how a green policy, however well-intentioned, can spark backlash if it ignores social equity. As Martin (2019) argues, “green policies must also address equality – or they will be rejected”. If climate mitigation measures are not adapted for vulnerable groups, they can lead to widespread protests and erode public support for the entire transition.

In addition to geographic disparities, the digital transition can deepen social exclusion along lines of class, age, education, and demography. Significant portions of the population lack basic digital skills or affordable internet access. In Europe, nearly half of people do not have basic digital abilities (Bost, 2024). This digital divide threatens to perpetuate social exclusion, as those without access or skills cannot fully participate in an increasingly digital society (Bost, 2024). Vulnerable groups, such as older people, low-income households, migrants, and people in remote areas, are most at risk (Bost, 2024). For example, if government and essential services move online, digitally excluded citizens may struggle to obtain services or even job opportunities. The benefits of digital transformation concentrate among the digitally literate and connected, while others are left further behind. As Bost (2024) noted, the digital divide exacerbates existing inequalities, demanding urgent attention to “ensure no one is left behind in the digital age”. If unaddressed, digital exclusion could map onto and worsen other social divides (urban vs rural, young vs old, skilled vs unskilled), thereby compounding inequality within and between European societies.

Income and household-level impacts are another equity dimension. The twin transition triggers shifts in prices and living costs; for instance, carbon pricing may make fossil-based energy and goods more expensive, and funding the transition might involve higher taxes. These costs tend to be regressive unless actively counterbalanced. A foresight report acknowledged that “lower- and middle-income households may bear the brunt of the twin transition as the costs of essential resources, goods and services (...) undergo substantial hikes” (Aloisi, 2025b). If green policies drive up electricity, heating, or fuel prices without adequate subsidies or income support, energy poverty could increase, hitting poor families the hardest. Likewise, if purchasing an electric car or renovating a home for energy efficiency is financially out of reach for many, the transition could create a two-tier society of those who can afford green alternatives and those who cannot. Such inequalities in costs and access not only raise moral and social justice issues but can also provoke resistance to climate policies among the economically struggling population.

The twin transition is fundamentally reshaping labour markets, creating new kinds of “green” and “digital” jobs while rendering some traditional jobs obsolete. This process creates hazards in the form of job displacement and skills mismatches. As Raza (2025) put it, “structural transformation inevitably creates winners and losers among workers, companies, and regions”. The key challenge is that the losses, and the people who bear them, may be quite different from those who enjoy the gains, leading to concentrated hardships.

In the green transition, industries that must shrink or transform include coal mining, oil & gas, peat extraction, and carbon-intensive manufacturing. The EU’s commitment to phase out coal, for example, directly threatens tens of thousands of jobs in mining regions of Poland, Germany, Czech Republic, Romania and beyond. If mines and coal plants close, not only miners but also entire communities dependent on those facilities face economic decline. The European Green Deal explicitly recognized this and established the Just Transition Mechanism (JTM) to provide funding and support to such regions. Yet, analysts argue the current JTM is “insufficient to mitigate adverse regional impacts and is scheduled to be phased out by 2026”, far too soon for the scale of transformation required (Raza, 2025). Coal and heavy industry regions

could see surging unemployment and young people leaving to find work elsewhere, resulting in “rust belt” style depopulation and distress (Sandmann et al., 2024). Moreover, if replacement jobs are not created rapidly, the social fabric of those communities may be threatened. A study of coal plant closures in Portugal noted significant local economic and social impacts from rapid decommissioning (Moreira et al., 2025). Thus, the timing and support of the green transition are critical: a poorly managed phase-out can leave workers and towns in a condition of “Death Valley” unemployment (Moreira et al., 2025).

Meanwhile, the digital transition’s impact on employment is translated into AI and automation displacing workers in various sectors. Advances in digitalisation can eliminate or significantly change jobs in manufacturing, retail, transportation, clerical works, and even skilled professions via algorithms. While digital transformation will create new jobs, those jobs often require different skills and are in different locations than the ones lost. Rapid automation can lead to structural unemployment for workers whose skills become outdated. Job displacement and digital divides are major risks accompanying twin transition; high-skilled expert work will be on demand in the long term, while blue-collar workers can be negatively affected in a very serious way (Alasoini, 2024).

Another potential hazard is the quality of jobs in the workers’ rights in the new economy. The digital transition has already given rise to the gig economy and algorithmic management practices that can erode traditional labour protections. Aloisi (2025a) points out that unregulated digital expansion can “threaten fundamental rights at work”. Scenarios such as AI use for hiring/firing decisions or 24/7 availability expectations show how digital tech could harm job quality even for those who remain employed.

Economically, one must also consider transitional costs and competitiveness issues. Investing in green infrastructures is expensive. Firms that cannot afford these investments or adapt in time may go out of business, especially SMEs, which could mean a reduction in competition and jobs in certain sectors. On the other hand, if Europe’s transition lags, there is a risk of industries moving production to regions with laxer climate rules, which can also cost jobs domestically. The EU tries to mitigate this by supporting innovations, but there are economic, social, and political risks that threaten to derail deployment of specific solutions (Raza, 2025).

Another set of hazards concerns the political realm. Transformative policies often entail short-term disruptions or perceived sacrifices for long-term gains. If these policies are not seen as fair or effective, they can lose legitimacy among the public. As the twin transition touches everyday aspects of life, it carries a political risk: policy resistance or populist backlash if people feel alienated or threatened by the changes.

Public opinion in Europe can erode quickly if policies are implemented in the wrong way. The Gilet Jaunes example discussed earlier is instructive: a single policy (fuel tax hike) sparked a movement that challenged not only that tax but broader governance, in part because it was seen as emblematic of elites imposing burdens on ordinary citizens (Martin, 2019). Ill-designed transition policies can become flashpoints for anger over inequality. Raza (2025) cautions that if welfare is cut and climate action stalls, it could fuel support for the populist right.

The twin transition is often portrayed as a win-win strategy for a sustainable and prosperous future. However, as demonstrated, research reveals a range of hazards and unintended consequences that must be addressed to fulfill the promises. Hazards include environmental side effects, unchecked digital revolution, rebound effects in climate policy, widening social inequalities between regions and demographics... Many scholars and researchers (Aloisi, 2025a; Martin, 2019) converge on the need to consider the social key issues of the twin transition to make it “socially desirable”. The twin transition is a complex journey with many trade-offs, and if the hazards outlined in this review are not managed, they could slow or derail progress towards Europe’s green and digital ambitions. Repeating European Commission’s (2022) words, the twin transitions “will be fair or will not be”.

## 5. Results from the semi-structured interviews with potentially vulnerable and disadvantaged cohorts

Assessing the potential negative impacts of the twin transition process on vulnerable and disadvantaged individuals requires a clear understanding of their perceptions and lived experiences. Semi-structured interviews are an effective qualitative research method to gather relevant insights from individuals who may be directly or indirectly affected by the green and digital transitions. Conducted as informal conversations led by experienced social science researchers, these interviews offer a valuable bottom-up perspective. This qualitative lens is essential for evaluating the adequacy and coherence of policy action plans aimed at advancing the twin transition, and for identifying critical opportunities to ensure that no one is left behind.

Between June and July 2025, semi-structured interviews were conducted with potentially vulnerable and disadvantaged individuals across six countries: Germany, Hungary, Ireland, Italy, Portugal and Spain. A total of 44 participants accepted the invitation of FITTER-EU partners to share their perceptions and personal experiences of how the twin transition has impacted their daily lives.

The key topics explored during the interviews, the ethical and methodological protocols followed, the participants' profiles, and a summary of the main findings are outlined below.

### ***Key discussion topics:***

- a) **Perceptions and awareness about the twin transition process and its meanings:** individual understanding of the twin transition process; perceived social (in)justice in the process; expected benefits and risks to personal wellbeing.
- b) **Personal experience through the lens of vulnerability:** direct and indirect impacts of the twin transition in personal, family, or household contexts; effects on daily routines and quality of life.
- c) **Coping mechanisms and adaptation:** availability of strategies, resources and tools used to address new challenges emerging from the transition process. Hopes, concerns, or expectations for the future.

### ***Ethical considerations and general principles of the recruitment of participants:***

Participation in the interviews was entirely voluntary, with participants free to decline or withdraw at any time. Researchers provided a plain-language explanation of the study's purpose, procedures, and confidentiality terms, along with an informed consent form. Verbal consent was also obtained before the interviews. All data were anonymized and stored securely, in compliance with data protection standards. Participants were primarily identified through trusted networks such as local NGOs, community organizations, social workers, support groups, and community centres, as well as through the personal contacts of social science researchers.

To ensure participant anonymity, each individual was assigned a unique code indicating only the corresponding national case study (DE, HU, IE, IT, ES, PT) and a sequential number. For example, DE 01 refers to participant number 1 from Germany, and IE 04 refers to participant number 4 from Ireland.

### ***Participant profiles:***

Following on the key outcomes of the ***D3.1 Report on vulnerabilities mapping under the baseline scenario***, the following profiles of potentially critical vulnerabilities were identified for each of the four economic sectors of FITTER-EU project across six national case-studies:

**Energy sector:** Low-income individuals (e.g. elderly, single parents, tenants, unemployed, people with disabilities or chronic illnesses) unable to afford energy-efficient technologies; low-skilled workers in the energy industry; workers laid off due to fossil fuel phase-out.

**Mobility and transport sector:** Low-income individuals unable to afford low-emission vehicles; residents in rural areas with limited access to affordable public transportation ("forced transport poverty"); people with disabilities who depend on personal vehicles for mobility; low-skilled or laid-off workers from the automotive sector.

**Housing and built environment sector:** Low-income individuals unable to afford energy-efficient home upgrades (climate insulation) or utility bills such as heating or air conditioning; residents in rural areas facing "forced housing poverty"; people with disabilities in need of adapted living spaces; individuals with chronic health conditions requiring consistent energy access (e.g. respiratory support); low-skilled and temporary workers in the construction or housing sectors; immigrants, refugees, and ethnic minorities.

**Agriculture sector:** small family farm owners; owners and managers of medium-sized agricultural enterprises; female entrepreneurs and workers in agriculture; low-skilled and seasonal agricultural workers.

#### ***Semi-structured interview protocol:***

Each interviewer received a common interview guide containing open-ended questions linked to the key discussion topics. Researchers were encouraged to adapt their language to ensure clarity, avoid jargon, and maintain cultural appropriateness based on the interviewee's background.

**Before the interview:** Interviewers confirmed informed consent (verbally and/or in writing), explained confidentiality measures, requested permission to record, and provided time for participants to ask questions.

**During the interview:** Researchers created a safe, respectful, and non-judgmental environment. They were encouraged to follow participants' leads where appropriate, remain empathetic, and stay alert to signs of discomfort, offering breaks or the option to stop the interview at any point.

**After the interview:** Interviewers thanked participants, reiterated the confidentiality of their input, and addressed any remaining questions or concerns.

#### ***Data analysis:***

This exploratory study employed a grounded theory methodology (Glaser & Strauss, 1967), which was appropriate for the qualitative nature of the research. This approach enabled the identification of themes and patterns that emerged from participants' insights into the twin transition process, their personal experiences of vulnerability, and the coping mechanisms used to address new challenges. Interview data were manually coded to capture recurring themes related to the three main discussion topics. Connections among these insights were then examined to refine and consolidate emerging themes, which were systematized and re-examined until data saturation was reached.

#### ***Main outcomes:***

The twin transition process has a multifaceted impact on the lives of individuals and families in the countries studied. Changes taking place as a consequence of new environmental policies aimed at

preventing climate change and a progressive digitisation of public administration and most daily experiences deeply affect the routines and quality of life of potentially vulnerable and disadvantaged cohorts, generating from additional economic burdens to feelings of fear, anxiety and self-exclusion. Among the most frequent themes emerged during the semi-structured interviews, the following themes require special attention.

### **1. Lack of clear understanding and awareness about the meaning of the twin transition process**

Most participants show a very superficial understanding of the twin transition process concept and its meaning. Some respondents perceive these changes as useful but expensive, while others don't engage at all due to age, technological fear, or living in rural settings. Many interviewees, especially older and low-income individuals, don't understand the meaning of the digital and green transitions or do not associate the two concepts:

*"I think I've heard it, but I don't really understand it" (IE 05).*

*"I have no idea what the terms mean" (DE 01, ES 02).*

*"No, I don't think I've actually heard of the digital transition in relation to climate change and the green transition before. I have heard the green transition, and the digital transition, but not about the two in relation to one another" (IE 01).*

*"Yeah, I would have kind of heard a bit about them, because we're quite small, we wouldn't really have much kind of technology on the farm anyway, like a lot, the work is just kind of done by hand, the old-fashioned way and then, then like, since we've had the farm, we've tried to do everything, as organic and stuff as possible anyway" (IE 08).*

Some participants even perceive them as "new nonsense that young people talk about" (HU 01). While the green transition mostly elicits comments about climate change and very occasionally about sustainability, the digital transition is commonly related to the use of the Internet, apps and smartphones. For some of them, the lack of understanding and /or awareness about the purpose of the transition process becomes a reason for a manifested scepticism:

*"It's what it seems we have to live because there are people who decide for us without being in our shoes, and they are not interested in our opinion" (ES 08).*

*"You know, I definitely do the best within my means, but yeah, I do think I do feel somewhat largely powerless to the in terms of the climate because it's also like I know I can do my best and you know, it's not that my contribution doesn't count, but it's a small drop in a very, very large bucket. You know? So like, yes, I do feel kind of powerless through all the changes" (IE 05).*

Extreme and atypical weather events are commonly recalled as the reference of climate change awareness. Frequent heatwaves, floods, draughts, longer summers, shorter winters with reduced snowfall and more dangerous storms are mentioned in relation to the green transition concept. Some participants recall "sorting waste, using less water or eco-friendly cleaning supplies" (HU 02), or the public challenges "don't use more plastic, use more like paper bags and stuff" (DE 02) as associated to the green transition topic.



The meaning of digital transition is mainly related to *“sending more emails than the mail and saving data but also I imagine it's like having a receipt of the shops like in the email and not wasting paper”* (DE 02), Internet, Facebook and online stuff.

Some participants highlight the impact of the digital transition on their daily life on social networking and employment opportunities. Only a few interviewees use online banking and pay their bills online, although the experience is often challenging for them:

*“I am a wheelchair user and I am on a couple of ethics committees. And before COVID, I would have to travel into these meetings in the city centre, either from my house here or from my parents' house in the countryside. So I could be travelling either four miles or 60 miles to attend these meetings. Now it's all done online with Teams and prior to that was Zoom meetings. So that has been very beneficial”* (IE 01).

*“Yes, everything is online, basically, yes, online doctor and everything with the bank also goes through the phone... What's the negative? I recently changed my phone, I had to go back to the bank to ask for the password, because I had no idea, of course, I was going with the fingerprint... I had forgotten, I no longer had it written down”* (ES 06).

Participants comment that they get information about climate change from public media, educational institutions, conversations with family and friends or through their professional training. However, information is not always clear, easy to understand, and accessible. Interviewees mainly recall advertising posters or some neighbourhood recycling activities. Some participants state that they try to be “eco-conscious” (HU 02) through recycling, using less water and eco-friendly cleaning products, buying second-hand products, and generally reducing waste and energy consumption:

*“Everything we're trying to do is to kind of with as little impact I suppose as possible, or a positive impact I suppose if it's making an impact on the environment and around just where we are, because we are so close to the water, you kind of have to be very conscious with that as well”* (IE 08).

Furthermore, the high utility bills make people “electricity cost-conscious” (IE 02), unplugging unnecessary appliances (HU 10) and trying to adapt their sleep patterns to ventilation patterns to manage the heat or cold in inefficient homes (DE 01). Some choose more affordable heating options such as wood, gas, or oil.

## **2. Increased financial hardships, anxiety and frustration**

Across several countries, vulnerable households are facing increasing financial pressure due to higher living costs and soaring utility bills. Furthermore, the high costs of innovative green and sustainable options make them automatically out of reach for this group. Purchasing electric vehicles, installing solar panels, heat pumps or insulating homes for energy efficiency is widely viewed as a luxury that only wealthier households can afford:

*“We still have to make a lot of sacrifices. Being up to date doesn't always mean you can afford to follow through fully”* (PT 10).

The feeling of housing insecurity is also widespread, especially among young low-income and single-family individuals. Most of the study participants are tenants of shared apartments or beneficiaries of social housing options. Finding housing is “very difficult” and “almost impossible” due to high costs, shortages, and discrimination by landlords (HU 08, DE 01, DE 02, ES 01, ES 10). Tenants often have “no power” to make

any energy-efficiency improvements (this information was found especially relevant in interviews in Germany and Ireland):

*“The current contract is also limited, but I had to search for a long time and had several interim tenancies/sublets before I found something longer-term again because there is simply far too little living space, far too much roof, far too many interested parties and also because landlords are discriminatory. It's precisely because I'm currently unemployed that some landlords didn't want me” (DE 01).*

Most interviewees live in old, poorly insulated houses. Problems of moisture, mold, physical barriers, lack of resources to keep their homes warm in winter and cool in summer are common across all countries included in the sample:

*“The house is already 40 years old. I have a wood burner, but I always have to find or buy firewood, and that's expensive. Sometimes by spring I've run out of firewood and I have to wrap up in blankets” (PT 04).*

*“I don't know the rating, but it's terribly cold and especially because it's directly affecting me — like how freezing it gets in the winter. I remember some days you could literally see your breath indoors. I'd be in bed with two pairs of pyjamas on, all my duvets and hot water bottles — and the likes — just trying to keep warm” (IE 05).*

*“We heat mostly with the fireplace. We have electric heaters, but only for the son's room” (PT 09).*

Poor energy efficiency in older homes and outdated but relatively cheap appliances often result in extremely high utility expenses:

*“I, for example, have an air conditioner that I've had for 28 years, and I have to be spending a lot on it, but I can't afford to buy another one. As long as it works, that's where I have it” (ES 07).*

In some cases, these costs are described as a “nightmare,” particularly for low-income families and pensioners, who struggle to cover basic energy needs (HU 08, DE 04, IE 05, ES 01, ES 02). Participants from Hungary, Spain, Ireland, and Germany highlight this challenge, with heating bills in Spain reaching up to €400 in a single winter month, and significantly increased costs in Germany and elsewhere:

*“We'd love to install solar panels, but it's not something we can afford just yet” (PT 08).*

In most countries included in the research sample, public transport system is widely described as inaccessible and inefficient in rural areas. In some countries, it is described as “useless” due to poor coverage or reliability (HU 01, HU 06), while in others it is effectively “non-existent” for people with reduced mobility (IE 01). The accessibility of public transport is a critical issue for participants with disabilities. While they highlight a higher number of accessible urban bus lines and metro stations, regional trains are still an issue:

*“They never really tell you if an accessible train is coming, and maybe if they do tell you that an accessible train is coming, it doesn't actually arrive because it seems they make last-minute changes to the tracks and any train can arrive, especially in commuter services and local trains” (ES 01).*

In certain cases, available services do not align with work schedules, making them impractical for daily commuting. As a result, many individuals are forced to rely on old diesel vehicles for daily commuting but

face rising fuel prices, the threat of future restrictions and bans on accessing low-emission zones, and additional taxes on combustion-engine vehicles, generating deep anxiety among those who cannot afford to replace their old cars. According to Spanish interviewees, new environmental policies (particularly low emission zones) and proposed restrictions on older diesel vehicles act as "*exclusion criteria*" that disproportionately affect people with limited incomes or special mobility needs, leaving them effectively "*stranded*" (ES 01, ES 08, ES 09).

Furthermore, people who lost their jobs in coal mines or other phasing-out related industries in Spain and Hungary, argue that the transition has caused a significant negative impact on their economic status, depopulation of their cities and grim employment prospects:

*"Here we have gone from having abundance to having nothing. And that contrast is very complicated, it is very difficult to digest. That has caused a lot of harm"* (ES 05).

### 3. Widening digital divide

The progressive digitization of public administration requiring the availability of digital ID and basic digital literacy to perform all procedures elicit frustration and feelings of anxiety, fear and helplessness among most of the interviewees. Furthermore, the ever-increasing replacement of in-person communication channels with online automatic tools and chatbots in banking, social welfare, health and education services make people feel "*alienated*" (HU 01), "*scared*" (ES 10) or "*stupid*" (HU 08) when faced with passwords or digital ID. Older and less digitally literate people find it "very difficult" to use so many platforms and remember passwords, or they feel "*overwhelmed and exposed to fraud and scams*" (HU 07, ES 01). For many participants digital procedures are commonly associated with increased bureaucracy and data privacy risks. In parallel, the lack of resources to purchase modern digital devices, to get broadband Internet and permanent mobile connections emphasizes the digital accessibility gap among low-income individuals, the elderly and residents in rural areas:

*"I still go pay the electricity bill at the shop in town"* (PT 07).

Some participants keep using traditional old phones just to call and receive calls, without any Internet connection. The support of family members and warm experts (friends, neighbours) to overcome digital barriers has been highlighted as one of the most useful coping mechanisms. Children and grandchildren often "*help with bills, manage digital procedures*", or teach how to use new technologies, smartphones, apps and devices (HU 01, ES 01, ES 10, PT 06, PT 04, PT 07):

*"Many things I have to do, I don't know how to do them. So I have to ask my grandchildren or my son-in-law. Because there are things I don't catch ... My family taught me how to do it several times and now I know how to use it and enter in the app"* (ES 10).

*"Well, when needed, I ask my children for help. I only answer calls. Learning new things like that isn't for me anymore"* (PT 04).

*"I have to help my grandparents a lot, whatever it may be, whether it's health issues or issues related to checking old files of theirs that are now digitized and that used to be done on paper and through phone calls, and now no one answers them. Basically, I have to do everything for them since they are not able to do that"* (ES 01).

*“The thing is that they keep coming out with new things, like Bizum, like paying with your mobile... we are modernizing. Of course, you have young kids who help you out, who teach you, but it’s complicated, especially for older people” (ES 06).*

The digital pressure is particularly critical in the workplace, where the reliance on digital control systems and automation generates fear and anxiety among low-skilled and older individuals of not being able to cope with new technologies and/or learn as fast as younger employees, which could lead to job loss and being forced permanently out of the workforce. This pressure adds to the feelings of insecurity and anxiety over the green transition, which has already resulted in significant job losses in certain sectors, particularly through the closure of coal mines in Hungary and Spain. These closures have had *“devastating”* economic consequences for local communities and have generated *“deep resentment”* and a strong sense of *“abandonment”* among former workers (HU 03, ES 05). Beyond the jobs that have already disappeared, there is a widespread fear of future unemployment among workers in fossil-fuel-dependent industries. Many express concern that their plants may be downsized or shut down entirely as a result of EU emissions regulations, which they fear would leave them without viable employment alternatives.

#### **4. Self- and system-driven exclusion from the process**

A perceived disconnect between the everyday needs of ordinary people and the personal interests of politicians and the wealthy was a recurring theme in participants’ narratives:

*“Nobody protects us, the honest workers” (HU 03).*

*“It’s a rented flat in which I have no decision-making powers. So I’m dependent on the landlord. In this case, I see responsibility on the part of the landlords, but also on the part of the state structures that are responsible for everything in the entire capitalist system, that there is no interest in sustainability, that the interest in making money prevails” (DE 01).*

Many participants expressed a strong sense of abandonment by their governments and a general disinterest in anything related to politics. They voiced scepticism about support programs, doubting such initiatives would ever benefit people like them. Although subsidies for solar panels, electric vehicles, and electricity vouchers exist, these technologies and energy-saving options were often perceived as inaccessible:

*“I understand that there are grants, but it is more expensive because you have to go through the companies that have agreements signed with the government, so it’s kind of like that, and they can charge you whatever they want” (IE 02).*

*“We hear people talk about it, but it feels far from our reality. Those things are for younger folks with better education and more money” (PT 06).*

Frustration was especially pronounced among participants from Spain and Hungary. Instead, they feel that the burdens of climate action, driven by what they see as the excessive consumption of the wealthy, are unfairly placed on the shoulders of the most vulnerable:

*“They are nonsense that are being proposed to be done, they don’t allow fishing for a trout in the river, but they do allow building a factory that dumps all the garbage into the river” (ES 08).*

*“And yes, that subsidies are provided. In other words, it's not sustainability that's being promoted, but what money can be made from. And that it's made super difficult for individuals if they don't have that much money, if they don't have their own house and don't have to worry about it” (DE 01).*

Adapting to digital and green transitions was also seen as a significant challenge. The steep learning curve and time investment required were often incompatible with the daily pressures participants faced in meeting their basic needs. This exacerbated feelings of exclusion, frustration, and fatigue:

*“I see that it really makes my grandparents feel very bad and makes them feel very excluded, and they cannot, I mean, even if they want to, they cannot pay attention to things that directly affect their private life” (ES 01).*

*“And considering there's like, especially in my house that I'm living in and there's a lot of other issues that are more at the forefront. So, like, again, I feel like solar panels might be in the end of a bit of a longer list, right?” (IE 05).*

*“Come on, either you manage to progress a little with new technologies, or a family member, a child, a nephew, whatever helps you out, or you're screwed. Nobody provides for you, nobody helps you with anything, and nobody gets you off the hook, this is clear. At least here” (ES 06).*

*“I can open a computer, open the mail but that's all, if I have to do anything I doubt and I don't do it because I'm scared and I don't know how to do things in the computer” (ES 10).*

*“If I have to go online or deal with paperwork, I won't do it” (PT 07).*

## Conclusions

The twin transition has a complex and often challenging impact on potentially vulnerable and disadvantaged individuals and their families, influencing their daily lives, financial stability, and emotional well-being. Many people, particularly older adults and those with lower levels of education, struggle to grasp the meaning and implications of these transitions. They are often perceived as abstract, confusing, or even exclusionary.

At the same time, rising energy costs, precarious housing conditions, and the prohibitive costs of new technologies place a significant financial strain on low- and middle-income households. Eco-friendly options such as electric vehicles or home energy renovations are frequently out of reach, exacerbating economic difficulties and generating anxiety about the future.

Digitalization introduces its own set of barriers, including limited access to devices or the internet, low digital literacy, and associated feelings of fear, anxiety, dependence, and exclusion. Older individuals and those with fewer skills are especially affected by the widening digital divide.

Concerns around public transportation are also widespread. It is often viewed as unreliable or inaccessible, particularly for those with limited mobility. Meanwhile, environmental policies such as low-emission zones are seen as disproportionately affecting people with fewer resources and relying on old cars and those living in remote or rural areas.

Compounding these challenges are broader issues such as job insecurity especially among low-skilled and in fossil fuel-dependent industries.

Overall, the transition process is widely perceived yet another source of anxiety, uncertainty, and financial hardship for those already facing vulnerability.

## 6. Results: Hazards, Disadvantaged Groups, and negative impacts

As stated before, this section synthesises evidence from literature review and semi-structured interviews, alongside the identification of vulnerable cohorts in D3.1 and the future scenarios developed in D3.2 (result of semi-structured interviews and a Delphi survey). It is important to recall that the Delphi survey provided a structured consensus-building process through which experts assessed the likelihood and impact of key trends and variables shaping the 2050 scenarios. Conducted in two iterative rounds, the survey engaged a diverse panel of experts across the four sectors under study (energy, mobility, housing & built environment, and agriculture), ensuring that the results reflect collective and cross-sectoral expertise rather than individual viewpoints.

In total, 62 experts participated in the Delphi survey, exceeding the target of 50. The largest share came from Spain (22), followed by Portugal (12) and Italy (10). By sector, the strongest representation was in energy (20 experts), followed by mobility & transport (19), agriculture (12) and housing & built environment (11). The recruitment strategy combined experts from the FITTER stakeholder database, snowball referrals and targeted invitations, ensuring disciplinary and geographical diversity.

The findings presented here highlight that hazards rarely occur in isolation. Instead, they interact, amplify one another and generate cumulative disadvantage. Central to this analysis is the recognition that hazards translate into tangible and often overlapping negative impacts that disproportionately affect disadvantaged groups, typically those already experiencing socioeconomic precarity.

Thus, the findings systematically link disadvantage groups to specific negative impacts through specific hazards, offering an understanding of where and how the risks of the twin transition are more acute. In addition to this, through the experts' consensus on the likelihood and severity of impact extracted from the Delphi Survey in D3.2, a risk index is calculated, which will be the base to establish the order of prioritization of the hazards.

Section 7 offers a complete summary table of all the hazards, negative impacts, Disadvantaged Groups, and risk index.

### 6.1 Energy

#### 6.1.1 Slow/Uneven grid upgrades and limited renewable infrastructure outside cities

##### Context and pathway to inequality

Slow or uneven grid upgrades and the limited deployment of renewable energy infrastructure outside major urban centres constitute a critical hazard, as they shape who can benefit from the decarbonization of the energy system and when. The energy transition requires not only cleaner generation but also modern and well-connected distribution networks capable of integrating decentralized renewable sources and supporting storage solutions while enabling demand-side management. When investment in these networks is delayed or concentrated in already advantaged urban areas, large segments of the population located in rural or remote regions are left with outdated infrastructure that cannot support affordable and clean energy access.

This uneven rollout leads to a two-speed transition, where cities gain early access to low-cost renewable energy and innovative grid services, while non-urban areas remain dependent on older and more expensive supplies. The absence of modernized grids also restricts the feasibility of local renewable projects and

prevents households from participating in decentralized energy markets or benefitting from dynamic pricing schemes. This has a cumulative effect: higher and more volatile energy costs persist in rural areas, and opportunities for cost savings or revenue generation through prosumer activities are missed, deepening territorial inequalities

### Disadvantaged groups and negative impacts

The groups most affected by this hazard are rural and remote residents, who face both structural and geographical barriers to energy transition benefits. These communities are often served by aging distribution lines with limited capacity to handle the variability of renewable generation or the integration of storage systems, meaning they cannot easily adopt or connect technologies such as rooftop solar panels. As a result, they remain reliant on centralized, fossil fuel-based generation and are exposed to higher per-unit energy costs, especially during periods of price volatility. The lack of grid upgrades in these areas also means that when innovative tariffs or local energy sharing schemes become available, rural households are excluded by default, unable to participate due to physical and technical constraints.

For households already struggling with low or unstable income, the inability to access cheaper and cleaner energy options exacerbates existing economic vulnerability, creating a cycle where higher costs reduce disposable income and limit investment capacity in energy efficiency measures. Over time, this gap between urban and rural energy access risks broadening social and economic disparities, reinforcing spatial inequality in both living standards and participation in the green transition.

### Prioritization

| Related Delphi statement: Households without solar panels will face higher long-term energy costs, exacerbating inequality |                   |                  |
|----------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                 | Number of answers | % over the total |
| Very high                                                                                                                  | 0                 | 0%               |
| High                                                                                                                       | 4                 | 50%              |
| Neutral                                                                                                                    | 1                 | 12.5%            |
| Low                                                                                                                        | 2                 | 25%              |
| Very low                                                                                                                   | 1                 | 12.5%            |
|                                                                                                                            |                   |                  |
| Impact                                                                                                                     | Number of answers | % over the total |
| Very high                                                                                                                  | 0                 | 0%               |
| High                                                                                                                       | 3                 | 37.5%            |
| Neutral                                                                                                                    | 3                 | 37.5%            |
| Low                                                                                                                        | 1                 | 12.5%            |
| Very low                                                                                                                   | 1                 | 12.5%            |
|                                                                                                                            |                   |                  |
| Averages                                                                                                                   | Likelihood        | Impact           |
|                                                                                                                            | 3                 | 3                |
|                                                                                                                            |                   |                  |
| Risk index                                                                                                                 | 9                 |                  |

### 6.1.2 Digital-by-default energy management without adequate support

#### Context and pathway to inequality

The shift toward digital-by-default energy management, exemplified by the widespread introduction of smart meters or app-based tariffs, presents a substantial hazard in the twin transition when it is implemented without adequate support for users with low digital literacy or limited access to technology. In theory, these digital tools enable more precise consumption monitoring and opportunities for cost savings through demand-side response. However, when their adoption is made mandatory or when analogue alternatives are phased out, those unable to navigate digital systems effectively are left at a disadvantage. The problem is exacerbated by the complexity of tariff structures and interfaces, which often require not only basic internet skills but also the ability to interpret technical consumption data and make time-sensitive decisions to benefit from lower rates.

Without accessible design and user-friendly support, the promise of smarter energy management turns into a barrier. Individuals may be unable to switch to more advantageous tariffs and miss out on incentive programmes, and they also might depend on intermediaries (such as family or friends) to manage their accounts). This dependence creates new risks, such as exposure to fraud or mismanagement of bills, and even a loss of privacy. Instead of improving empowerment and efficiency, poorly supported digitalization can thus deepen inequality and reduce the very consumer autonomy it is meant to enhance.

#### Disadvantaged groups and negative impacts

The groups most affected by this hazard are elderly people and individuals with limited digital literacy, and those without reliable internet access or appropriate devices. Older adults, and people with low digital literacy, may struggle with the rapid shift to app-based systems due to unfamiliarity with digital interfaces or a preference for in-person communication, leading to confusion and errors in account management. People in low-income households may lack technological infrastructure, making continuous digital access to energy services impractical or impossible. For these disadvantaged groups, the negative impacts include paying higher bills because they cannot take advantage of time-of-use or dynamic tariffs, being excluded from automated savings mechanisms, and enduring stress or anxiety about navigating complex systems. In some cases, the reliance on intermediaries to manage accounts introduces a dependency that can erode personal control over financial decisions, heightening vulnerability to scams.

#### Prioritization

| Related Delphi Statement 1: Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumers rights |                   |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                                      | Number of answers | % over the total |
| Very high                                                                                                                                                       | 3                 | 37.5%            |
| High                                                                                                                                                            | 3                 | 37.5%            |
| Neutral                                                                                                                                                         | 2                 | 25%              |
| Low                                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                                 |                   |                  |
| Impact                                                                                                                                                          | Number of answers | % over the total |
| Very high                                                                                                                                                       | 0                 | 0%               |
| High                                                                                                                                                            | 5                 | 62.5%            |
| Neutral                                                                                                                                                         | 3                 | 37.5%            |
| Low                                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                                        | 0                 | 0%               |



|                     |                   |               |
|---------------------|-------------------|---------------|
|                     |                   |               |
| <b>Averages</b>     | <b>Likelihood</b> | <b>Impact</b> |
|                     | 4.125             | 3.625         |
|                     |                   |               |
| <b>Risk index 1</b> | 14.95             |               |

| <b>Related Delphi Statement 2: Energy literacy will become essential for consumers to actively participate in dynamic pricing schemes and smart grid programs</b> |                          |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                                 | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                  | 2                        | 25%                     |
| <b>High</b>                                                                                                                                                       | 4                        | 50%                     |
| <b>Neutral</b>                                                                                                                                                    | 2                        | 25%                     |
| <b>Low</b>                                                                                                                                                        | 0                        | 0%                      |
| <b>Very low</b>                                                                                                                                                   | 0                        | 0%                      |
|                                                                                                                                                                   |                          |                         |
| <b>Impact</b>                                                                                                                                                     | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                  | 2                        | 25%                     |
| <b>High</b>                                                                                                                                                       | 2                        | 25%                     |
| <b>Neutral</b>                                                                                                                                                    | 3                        | 32.5%                   |
| <b>Low</b>                                                                                                                                                        | 1                        | 12.5%                   |
| <b>Very low</b>                                                                                                                                                   | 0                        | 0%                      |
|                                                                                                                                                                   |                          |                         |
| <b>Averages</b>                                                                                                                                                   | <b>Likelihood</b>        | <b>Impact</b>           |
|                                                                                                                                                                   | 4.5                      | 3.625                   |
|                                                                                                                                                                   |                          |                         |
| <b>Risk index 2</b>                                                                                                                                               | 16.31                    |                         |

**Total average risk index** = 15.63

### 6.1.3 Abrupt decarbonization closures in fossil-fuel/energy value chains

#### Context and pathway to inequality

The abrupt closure of fossil fuel-based facilities and related activities in the energy value chain, driven by decarbonization policies and the rapid deployment of low-carbon technologies, constitutes a significant hazard in the twin transition. While the green transition requires phasing out coal, oil, and gas production and associated industries, the speed and manner in which these closures occur can produce several social and economic disruptions if not accompanied by robust measures. Communities historically dependent on these sectors often face a sudden loss of economic anchors, as plants or mines shut down.

The resulting job losses are not easily absorbed by the local labour market, especially in regions where alternative industries are scarce and where skills are closely tied to the declining sector. The effects impact direct employees and contractors and supply chain firms whose livelihoods depend on the economic activity generated by these industries. In the absence of proactive reskilling programmes and targeted social support, as well as economic diversification strategies, the closure of carbon-intensive operations can trigger long-term regional decline and a sense of abandonment among affected communities.

### Disadvantaged groups and negative impacts

The disadvantaged groups most directly affected by this hazard are low-skilled energy sector workers, many of whom have built their careers in fossil fuel industries with limited opportunities to transition into equivalent employment. Older workers in these sectors are particularly vulnerable, as the prospect of retraining for a new profession late in their careers is often unrealistic, making early retirement on reduced income a likely outcome.

In communities where fossil fuel operations have been a dominant employer for decades, the impact extends to entire families and local economies. Job losses reduce household income, leading to increased financial insecurity and higher risks of poverty. Local businesses suffer from reduced demand, and public revenues decline, affecting the provision of infrastructure and social services. Younger generations may leave in search of better prospects elsewhere, accelerating demographic decline and weakening social cohesion. This hazard undermines the economic fabric of their communities and limits their participation in the opportunities created by the green transition.

### Prioritization

| Related Delphi Statement 1: Low-skilled workers in extractive industries (coal, oil and gas sectors) will face job losses |                   |                  |
|---------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                | Number of answers | % over the total |
| Very high                                                                                                                 | 1                 | 12.5%            |
| High                                                                                                                      | 3                 | 37.5%            |
| Neutral                                                                                                                   | 3                 | 37.5%            |
| Low                                                                                                                       | 1                 | 12.5%            |
| Very low                                                                                                                  | 0                 | 0%               |
|                                                                                                                           |                   |                  |
| Impact                                                                                                                    | Number of answers | % over the total |
| Very high                                                                                                                 | 0                 | 0%               |
| High                                                                                                                      | 3                 | 37.5%            |
| Neutral                                                                                                                   | 5                 | 62.5%            |
| Low                                                                                                                       | 0                 | 0%               |
| Very low                                                                                                                  | 0                 | 0%               |
|                                                                                                                           |                   |                  |
| Averages                                                                                                                  | Likelihood        | Impact           |
|                                                                                                                           | 3.5               | 3.375            |
|                                                                                                                           |                   |                  |
| Risk index 1                                                                                                              | 11.8125           |                  |

| Fossil fuel phase-out policies will generate long-term regional unemployment risk in areas historically dependent on extractive industries |                   |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                 | Number of answers | % over the total |
| Very high                                                                                                                                  | 1                 | 12.5%            |
| High                                                                                                                                       | 2                 | 25%              |
| Neutral                                                                                                                                    | 3                 | 37.5%            |
| Low                                                                                                                                        | 2                 | 25%              |
| Very low                                                                                                                                   | 0                 | 0%               |
|                                                                                                                                            |                   |                  |
| Impact                                                                                                                                     | Number of answers | % over the total |
| Very high                                                                                                                                  | 0                 | 0%               |

|              |            |        |
|--------------|------------|--------|
| High         | 3          | 37.5%  |
| Neutral      | 5          | 62.5%  |
| Low          | 0          | 0%     |
| Very low     | 0          | 0%     |
|              |            |        |
| Averages     | Likelihood | Impact |
|              | 3.25       | 3.375  |
|              |            |        |
| Risk index 1 | 11.8125    |        |

**Total average risk index** = 11.8125

## 6.2 Housing and built environment

### 6.2.1 Landlord-tenant split incentives in retrofits

#### Context and pathway to inequality

The misalignment of incentives between landlords and tenants in relation to energy efficiency retrofits is a well-documented structural barrier that becomes a pronounced hazard in the context of the twin transition. Achieving climate neutrality in the building sector requires substantial investment in efficiency upgrades such as insulation or heating systems. However, when tenants are responsible for paying energy bills but do not own the property, they have little capacity or authority to initiate improvements. Landlords, for their part, may be reluctant to invest in retrofits when they do not directly benefit from the resulting lower energy costs, particularly if the payback period is long or uncertain. This split-incentive problem is further intensified when public subsidies or renovation schemes are designed in ways that favour homeowners over renters, or when eligibility requirements implicitly exclude tenants.

As a result, a large share of the rental housing stock remains energy inefficient, locking occupants into high consumption and elevated costs. In the absence of targeted measures, the hazard perpetuates a cycle in which tenants in poorly performing buildings cannot access the benefits of reduced bills or improved comfort, and landlords lack motivation to change the situation, even when the broader policy environment pushes for a more efficient housing stock.

#### Disadvantaged groups and negative impacts

The disadvantaged groups most affected by this hazard are low- and moderate-income tenants, especially those living in older or substandard rental properties. These households are structurally dependent on their landlords' willingness to undertake renovations, and in many cases the owners may be unwilling or financially unable to do so. The result is that tenants continue to pay disproportionately high energy bills for inadequate heating or cooling, while also experiencing poor indoor environmental quality.

This situation can lead to a range of negative impacts such as higher rates of energy poverty and reduced disposable income for other essential needs. For families with children, the inability to maintain a stable indoor climate can affect wellbeing and learning conditions. In many urban contexts, rental markets are tight, making relocation to a more energy-efficient home unaffordable or impossible. Consequently, the landlord-tenant split incentive does not only delay progress toward climate and energy goals but also reinforces inequalities in housing quality and affordability, creating a persistent and hard-to-break link between tenure status and vulnerability in the twin transition.

## Prioritization

| Related Delphi Statement 1: Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households |                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                      | Number of answers | % over the total |
| Very high                                                                                                                                       | 1                 | 20%              |
| High                                                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                                                         | 1                 | 20%              |
| Low                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                 |                   |                  |
| Impact                                                                                                                                          | Number of answers | % over the total |
| Very high                                                                                                                                       | 1                 | 20%              |
| High                                                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                                                         | 1                 | 20%              |
| Low                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                 |                   |                  |
| Averages                                                                                                                                        | Likelihood        | Impact           |
|                                                                                                                                                 | 4                 | 4                |
|                                                                                                                                                 |                   |                  |
| Risk index 1                                                                                                                                    | 16                |                  |

| Related Delphi Statement 2: Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050 |                   |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                                                 | Number of answers | % over the total |
| Very high                                                                                                                                                                  | 3                 | 60%              |
| High                                                                                                                                                                       | 2                 | 40%              |
| Neutral                                                                                                                                                                    | 0                 | 0%               |
| Low                                                                                                                                                                        | 0                 | 0%               |
| Very low                                                                                                                                                                   | 0                 | 0%               |
|                                                                                                                                                                            |                   |                  |
| Impact                                                                                                                                                                     | Number of answers | % over the total |
| Very high                                                                                                                                                                  | 2                 | 40%              |
| High                                                                                                                                                                       | 2                 | 40%              |
| Neutral                                                                                                                                                                    | 0                 | 0%               |
| Low                                                                                                                                                                        | 1                 | 20%              |
| Very low                                                                                                                                                                   | 0                 | 0%               |
|                                                                                                                                                                            |                   |                  |
| Averages                                                                                                                                                                   | Likelihood        | Impact           |
|                                                                                                                                                                            | 4.6               | 4                |
|                                                                                                                                                                            |                   |                  |
| Risk index 2                                                                                                                                                               | 18.4              |                  |

**Total average risk index** = 17.2

### 6.2.2 High adaptation costs for homeowners amid price-sensitive incentives

#### Context and pathway to inequality

Achieving climate targets require extensive retrofitting of the housing stocks: improving insulation, replacing inefficient heating and cooling systems, integrating renewable energy solutions, implementing measures to withstand extreme weather... Public subsidies and incentive programmes exist, but they often cover only part of the total cost, leaving homeowners responsible for a substantial share of the investment.

For households with limited financial capacity, even a partially subsidized retrofit can represent an unmanageable expense, particularly when upfront payments are required before reimbursement. Additionally, fluctuating energy prices can influence the perceived return on investment, with some homeowners delaying or abandoning upgrades if anticipated savings are uncertain.

### Disadvantaged groups and negative impacts

The disadvantaged groups affected by this hazard are low-income homeowners, many of whom live in older, less efficient properties that require the most extensive and costly renovations. For these households, the gap between the subsidy amount and the actual cost is often too high, leading to incomplete retrofits or the postponement of necessary works. Such outcomes can perpetuate high energy bills and increased exposure to climate-related risks such as heatwaves.

The financial strain associated with attempting these upgrades can also lead to increased debt and reduced capacity to cover other essential expenses. In areas where climate impacts are intensifying, the inability to invest in resilience measures can result in tangible harm to health and safety, with vulnerable occupants more exposed to climate hazards. This hazard can reinforce the divide between households able to invest in high-standard, climate-resilient homes and those trapped in substandard housing.

### Prioritization

| Related Delphi Statement 1: Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households |                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                      | Number of answers | % over the total |
| Very high                                                                                                                                       | 1                 | 20%              |
| High                                                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                                                         | 1                 | 20%              |
| Low                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                 |                   |                  |
| Impact                                                                                                                                          | Number of answers | % over the total |
| Very high                                                                                                                                       | 1                 | 20%              |
| High                                                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                                                         | 1                 | 20%              |
| Low                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                 |                   |                  |
| Averages                                                                                                                                        | Likelihood        | Impact           |
|                                                                                                                                                 | 4                 | 4                |
|                                                                                                                                                 |                   |                  |
| Risk index 1                                                                                                                                    | 16                |                  |

| Related Delphi Statement 2: Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050 |                   |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                                                 | Number of answers | % over the total |
| Very high                                                                                                                                                                  | 3                 | 60%              |

|              |                   |                  |
|--------------|-------------------|------------------|
| High         | 2                 | 40%              |
| Neutral      | 0                 | 0%               |
| Low          | 0                 | 0%               |
| Very low     | 0                 | 0%               |
|              |                   |                  |
| Impact       | Number of answers | % over the total |
| Very high    | 2                 | 40%              |
| High         | 2                 | 40%              |
| Neutral      | 0                 | 0%               |
| Low          | 1                 | 20%              |
| Very low     | 0                 | 0%               |
|              |                   |                  |
| Averages     | Likelihood        | Impact           |
|              | 4.6               | 4                |
|              |                   |                  |
| Risk index 2 | 18.4              |                  |

**Total average risk index** = 17.2

### 6.2.3 Exclusion of migrants/refugees/undocumented people from housing subsidies or formal programmes

#### Context and pathway to inequality

Many public schemes designed to improve the energy efficiency and climate resilience of houses and apartments have eligibility criteria tied to factors such as citizenship (or legal residency status), or formal property ownership. These criteria, often combined with administrative complexity and language barriers, systematically prevent certain population groups from accessing support.

As a result, migrants and refugees are disproportionately likely to live in substandard or poorly insulated housing that is more vulnerable to extreme temperatures and energy price volatility. The hazard is exacerbated by the fact that these groups frequently occupy rental properties in the informal sector, where landlords may have little incentive to invest in improvements or to participate in formal subsidy schemes.

#### Disadvantaged groups and negative impacts

The disadvantaged groups most affected are migrants, refugees, and undocumented residents, who often already face intersecting vulnerabilities related to income insecurity and precarious employment. Exclusion from renovation and subsidy programmes means they remain in housing with high energy demand and limited protection from climatic extremes, exposing them to elevated health risks and persistent energy poverty; this can be exacerbated by overcrowding.

The lack of official tenancy agreements in many cases also prevents occupants from asserting their rights or negotiating improvements, locking them into a cycle of poor-quality housing and high operating costs. This perpetuates socio-economic disadvantage and undermines broader climate and energy objectives by leaving a segment of the housing stock (often in urban areas with already high social needs) outside of the transition process, which can reinforce both social exclusion and spatial inequality.

## Prioritization

| Related Delphi Statement 1: Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households |                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                      | Number of answers | % over the total |
| Very high                                                                                                                                       | 1                 | 20%              |
| High                                                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                                                         | 1                 | 20%              |
| Low                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                 |                   |                  |
| Impact                                                                                                                                          | Number of answers | % over the total |
| Very high                                                                                                                                       | 1                 | 20%              |
| High                                                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                                                         | 1                 | 20%              |
| Low                                                                                                                                             | 0                 | 0%               |
| Very low                                                                                                                                        | 0                 | 0%               |
|                                                                                                                                                 |                   |                  |
| Averages                                                                                                                                        | Likelihood        | Impact           |
|                                                                                                                                                 | 4                 | 4                |
|                                                                                                                                                 |                   |                  |
| Risk index 1                                                                                                                                    | 16                |                  |

| Related Delphi Statement 2: Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050 |                   |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                                                                 | Number of answers | % over the total |
| Very high                                                                                                                                                                  | 3                 | 60%              |
| High                                                                                                                                                                       | 2                 | 40%              |
| Neutral                                                                                                                                                                    | 0                 | 0%               |
| Low                                                                                                                                                                        | 0                 | 0%               |
| Very low                                                                                                                                                                   | 0                 | 0%               |
|                                                                                                                                                                            |                   |                  |
| Impact                                                                                                                                                                     | Number of answers | % over the total |
| Very high                                                                                                                                                                  | 2                 | 40%              |
| High                                                                                                                                                                       | 2                 | 40%              |
| Neutral                                                                                                                                                                    | 0                 | 0%               |
| Low                                                                                                                                                                        | 1                 | 20%              |
| Very low                                                                                                                                                                   | 0                 | 0%               |
|                                                                                                                                                                            |                   |                  |
| Averages                                                                                                                                                                   | Likelihood        | Impact           |
|                                                                                                                                                                            | 4.6               | 4                |
|                                                                                                                                                                            |                   |                  |
| Risk index 2                                                                                                                                                               | 18.4              |                  |

**Total average risk index** = 17.2

## 6.2.4 Digitalisation/automation of construction processes displacing low-skilled labour

### Context and pathway to inequality

Within the twin transition context, the construction sector is expected to play a pivotal role in delivering energy-efficient buildings and infrastructure adapted to climate change. But as digital tools such as prefabrication technologies and AI-supported planning become standard, traditional manual and on-site roles are being reduced or transformed. Tasks once performed by larger on-site teams can now be carried out by smaller, more technically skilled crews, often supported by off-site automated production.

These chances can enhance productivity and even reduce environmental impact, but they risk excluding workers whose experience lies primarily in manual, non-digital construction work. If the pace of technological adoption goes faster than the provision of tailored, accessible training programs, these workers may find themselves unable to adapt, resulting in job losses or transitions into lower-paid and less secure operations.

### Disadvantaged groups and negative impacts

Low-skilled construction workers, many of whom have built their livelihoods on manual trades and site-based labours, is the group most affected by this hazard. These workers often face barriers to accessing formal training (including cost or time constraints). In some cases, training opportunities are concentrated in urban centres, making them inaccessible to those in rural or remote areas where travel adds an additional burden.

The transition to digital and automatized construction can result in loss of income stability and reduced access to benefits linked to permanent employment contracts. For older workers, the challenge is even greater, as learning entirely new digital systems late in a career may feel unattainable. The economic and social impacts extend beyond individuals, affecting households dependent on this income and communities where construction is a major source of local employment. This hazard can create a segmented labour market in the construction sector, with a smaller core of digitally skilled workers benefitting from stable, well-paid jobs, and a growing periphery of displaced or causal workers with limited prospects in the green and digital economy.

| Related Delphi Statement 1: By 2050, the construction sector will be dominated by assembly and automation roles |                   |                  |
|-----------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                      | Number of answers | % over the total |
| Very high                                                                                                       | 1                 | 20%              |
| High                                                                                                            | 3                 | 60%              |
| Neutral                                                                                                         | 1                 | 20%              |
| Low                                                                                                             | 0                 | 0%               |
| Very low                                                                                                        | 0                 | 0%               |
|                                                                                                                 |                   |                  |
| Impact                                                                                                          | Number of answers | % over the total |
| Very high                                                                                                       | 1                 | 20%              |
| High                                                                                                            | 2                 | 40%              |
| Neutral                                                                                                         | 2                 | 40%              |
| Low                                                                                                             | 0                 | 0%               |
| Very low                                                                                                        | 0                 | 0%               |
|                                                                                                                 |                   |                  |
| Averages                                                                                                        | Likelihood        | Impact           |
|                                                                                                                 | 4.4               | 3.8              |



|                     |       |
|---------------------|-------|
| <b>Risk index 1</b> | 16.72 |
|---------------------|-------|

| <b>Related Delphi Statement 2: Retrofitting practices will increasingly rely on prefabricated and modular construction techniques to reduce costs and speed implementation</b> |                          |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                                              | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                               | 2                        | 40%                     |
| <b>High</b>                                                                                                                                                                    | 1                        | 20%                     |
| <b>Neutral</b>                                                                                                                                                                 | 0                        | 0%                      |
| <b>Low</b>                                                                                                                                                                     | 2                        | 40%                     |
| <b>Very low</b>                                                                                                                                                                | 0                        | 0%                      |
|                                                                                                                                                                                |                          |                         |
| <b>Impact</b>                                                                                                                                                                  | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                               | 2                        | 40%                     |
| <b>High</b>                                                                                                                                                                    | 2                        | 40%                     |
| <b>Neutral</b>                                                                                                                                                                 | 0                        | 0%                      |
| <b>Low</b>                                                                                                                                                                     | 1                        | 20%                     |
| <b>Very low</b>                                                                                                                                                                | 0                        | 0%                      |
|                                                                                                                                                                                |                          |                         |
| <b>Averages</b>                                                                                                                                                                | <b>Likelihood</b>        | <b>Impact</b>           |
|                                                                                                                                                                                | 3.6                      | 4                       |
|                                                                                                                                                                                |                          |                         |
| <b>Risk index 2</b>                                                                                                                                                            | 14.4                     |                         |

**Total average risk index** = 15.56

## 6.3 Agriculture and food

### 6.3.1 Increase of precariousness in seasonal/migrant agricultural work

#### Context and pathway to inequality

The agricultural sector has long relied on seasonal and migrant labour to meet peaks in demand, particularly for harvesting, but these workers are frequently employed under conditions that are poorly regulated and insecure. As agricultural systems are reshaped by the demands of decarbonization and digitalization, there is a risk that these workers remain excluded from the benefits of improved practices while continuing to bear the costs of instability.

In many cases, innovations in agri-tech or compliance with environmental requirements increase pressure on producers to reduce labour costs, which can exacerbate the reliance on precarious employment. Also, their work may be substituted and automatized, excluding them from the possibility to access even a temporary job. The absence of strong social protection measures for these workers creates a situation in which the transition to more sustainable food systems is achieved at the expense of those whose labour is crucial for the sector.

#### Disadvantaged groups and negative impacts

As stated before, seasonal and migrant farm workers, many of whom face intersecting disadvantages, are the disadvantaged groups most affected by this hazard. For these groups, the negative impacts include unstable employment patterns, low wages, unsafe or unhealthy working conditions, and limited access to

healthcare or social services. Migrant workers may avoid reporting exploitation or unsafe conditions for fear of losing their employment or residency status, perpetuating cycles of abuse. The persistence of these patterns can lead to the concentration of the benefits of innovation and sustainability among producers and consumers, while the costs are disproportionately borne by workers at the bottom of the value chain.

### Prioritization

| <b>Related Delphi Statement: Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labour</b> |                          |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                                        | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                         | 2                        | 50%                     |
| <b>High</b>                                                                                                                                                              | 2                        | 50%                     |
| <b>Neutral</b>                                                                                                                                                           | 0                        | 0%                      |
| <b>Low</b>                                                                                                                                                               | 0                        | 0%                      |
| <b>Very low</b>                                                                                                                                                          | 0                        | 0%                      |
|                                                                                                                                                                          |                          |                         |
| <b>Impact</b>                                                                                                                                                            | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                         | 4                        | 100%                    |
| <b>High</b>                                                                                                                                                              | 0                        | 0%                      |
| <b>Neutral</b>                                                                                                                                                           | 0                        | 0%                      |
| <b>Low</b>                                                                                                                                                               | 0                        | 0%                      |
| <b>Very low</b>                                                                                                                                                          | 0                        | 0%                      |
|                                                                                                                                                                          |                          |                         |
| <b>Averages</b>                                                                                                                                                          | <b>Likelihood</b>        | <b>Impact</b>           |
|                                                                                                                                                                          | 4.5                      | 5                       |
|                                                                                                                                                                          |                          |                         |
| <b>Risk index</b>                                                                                                                                                        | 22.5                     |                         |

### 6.3.2 Capital-intensive agri-tech that small producers can't finance

#### Context and pathway to inequality

The agricultural sector is under pressure to decarbonize and comply with stricter environmental and quality regulations, while simultaneously integrating digital technologies such as precision farming tools and automated machinery. Although these innovations can optimize resource use and improve market competitiveness, they often require substantial upfront investment in equipment and software. Smaller producers, particularly those with limited access to credit or with narrow profit margins, struggle to mobilise the necessary capital.

At the same time, meeting new sustainability certification requirements or compliance standards can mean additional work, further increasing costs. When the pace of technological and regulatory change exceeds the capacity of smaller farms to adapt, these hazards emerge as a process of exclusion from value chain, with the benefits of innovation being received disproportionately to larger, better-capitalised agribusiness.

#### Disadvantaged groups and negative impacts

The most affected are small and medium farmers. Limited financial resources restrict their ability to purchase modern machinery or adopt precision agriculture tools. Young farmers may lack the capital to modernize quickly enough to remain competitive, while older farmers may be less inclined or able to retrain in digital competencies.

The negative impacts include declining profitability and competitiveness. As consolidation accelerates, rural communities lose not only economic diversity but also social cohesion, as independent family farms are replaced by larger, more centralized operations, weakening the socio-economic fabric of regions dependent on small-scale agriculture.

#### Prioritization

| <b>Related Delphi Statement: No farm under 10 ha will be able to adopt precision technologies without joining a formal support network of service providers or cooperatives</b> |                          |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                                               | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                                | 3                        | 75%                     |
| <b>High</b>                                                                                                                                                                     | 0                        | 0%                      |
| <b>Neutral</b>                                                                                                                                                                  | 0                        | 0%                      |
| <b>Low</b>                                                                                                                                                                      | 1                        | 25%                     |
| <b>Very low</b>                                                                                                                                                                 | 0                        | 0%                      |
|                                                                                                                                                                                 |                          |                         |
| <b>Impact</b>                                                                                                                                                                   | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                                                | 2                        | 50%                     |
| <b>High</b>                                                                                                                                                                     | 1                        | 25%                     |
| <b>Neutral</b>                                                                                                                                                                  | 1                        | 25%                     |
| <b>Low</b>                                                                                                                                                                      | 0                        | 0%                      |
| <b>Very low</b>                                                                                                                                                                 | 0                        | 0%                      |
|                                                                                                                                                                                 |                          |                         |
| <b>Averages</b>                                                                                                                                                                 | <b>Likelihood</b>        | <b>Impact</b>           |
|                                                                                                                                                                                 | 4.25                     | 4.25                    |
|                                                                                                                                                                                 |                          |                         |
| <b>Risk index</b>                                                                                                                                                               | 18.0625                  |                         |

### 6.3.3 Sustainability standards that small-scale farmers and producers are unable to comply with

#### Context and pathway to inequality

European and national policies are increasingly requiring farmers to comply with stricter environmental and climate-related regulations, covering areas such as pesticide and fertilizer use, water consumption, soil protection, biodiversity conservation, and carbon footprint reduction. These standards (although they aim to make food systems more resilient and environmentally sustainable) often introduce additional costs and administrative requirements and obligations for farmers.

Compliance may involve investment in monitoring equipment or certification processes, or the adoption of alternative practices that can reduce short-term yields. Larger and more capitalized agribusinesses are generally able to adapt to these changes, spreading costs across bigger operations or dedicating staff to compliance tasks. Smaller farms, however, struggle to absorb the costs and administrative burdens, especially when multiple overlapping certifications are required to access different markets. This creates a structural divide in which the very standards designed to raise environmental performance risk marginalizing small producers and accelerating sectoral consolidation.

#### Disadvantaged groups and negative impacts

The disadvantaged groups most affected are small-scale farmers and producers with limited administrative capacity, who often lack both financial and human resources to comply with increasingly complex

sustainability requirements. For these farmers, the negative impacts include shrinking profit margins due to the additional costs of compliance and an increased vulnerability to losing contracts with distributors or retailers that impose strict sourcing criteria. As certification and monitoring systems often require the use of digital platforms for record-keeping and data submission, older farmers or those with limited digital literacy may be disproportionately affected.

This hazard can lead to farm closures, reducing diversity in the agricultural sector, and leading to the concentration of production in fewer, larger companies. The consequences of this can mean the weakening of rural economies, limiting the capacity of local food systems to provide stable and inclusive sources of employment and nutrition.

#### Prioritization

| Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems |                   |                  |
|---------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                          | Number of answers | % over the total |
| Very high                                                                                                           | 3                 | 75%              |
| High                                                                                                                | 1                 | 25%              |
| Neutral                                                                                                             | 0                 | 0%               |
| Low                                                                                                                 | 0                 | 0%               |
| Very low                                                                                                            | 0                 | 0%               |
|                                                                                                                     |                   |                  |
| Impact                                                                                                              | Number of answers | % over the total |
| Very high                                                                                                           | 2                 | 50%              |
| High                                                                                                                | 2                 | 50%              |
| Neutral                                                                                                             | 0                 | 0%               |
| Low                                                                                                                 | 0                 | 0%               |
| Very low                                                                                                            | 0                 | 0%               |
|                                                                                                                     |                   |                  |
| Averages                                                                                                            | Likelihood        | Impact           |
|                                                                                                                     | 4.75              | 4.5              |
|                                                                                                                     |                   |                  |
| Risk index                                                                                                          | 21.375            |                  |

## 6.4 Mobility and transport

### 6.4.1 High upfront cost of Electric Vehicles and complex/limited subsidies

#### Context and pathway to inequality

Electrification of private transport is a cornerstone of decarbonization strategies, intended to reduce emissions and improve air quality, particularly in urban areas. However, the purchase price of electric vehicles remains substantially higher than that of conventional internal combustion engine models, even when long-term operating costs are lower. Public incentive programmes, while designed to reduce this price gap, often require buyers to go across complex application procedures or provide significant funds upfront before reimbursement; for many households, these conditions create prohibitive financial and administrative barriers.

Consequently, the benefits of a cleaner, cheaper-to-run mobility go to higher-income individuals who can afford the initial investment and absorb the waiting period for subsidies. Meanwhile, households unable

to participate continue to rely on older and more polluting vehicles that incur higher running and maintenance costs, locking them into a cycle of transport-related disadvantage.

### Disadvantaged groups and negative impacts

The disadvantaged groups most affected by this hazard are low-income households and individuals with limited access to credit or savings, for whom the upfront capital required to purchase an electric vehicle is out of reach. Even when subsidies exist, the need to pay the full amount in advance or go through bureaucratic processes can deter or exclude those without financial buffers or bureaucratic literacy.

These households often rely on ageing vehicles with high fuel consumption, leading to disproportionate spending on fuel and repairs relative to their income. Over time, this reliance increases exposure to restrictions in urban areas where low-emission zones penalize or ban older vehicles, creating further economic and social barriers to accessing jobs, education, healthcare, and other essential services. In rural areas, where public transport alternatives are limited, the inability to transition to electric vehicles reinforces dependence on inefficient vehicles and leaves households vulnerable to fuel price volatility as well.

| Related Delphi Statement: Fiscal incentives for EVs will be redesigned to better target low-income households |                   |                  |
|---------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                    | Number of answers | % over the total |
| Very high                                                                                                     | 0                 | 0%               |
| High                                                                                                          | 2                 | 25%              |
| Neutral                                                                                                       | 4                 | 50%              |
| Low                                                                                                           | 2                 | 25%              |
| Very low                                                                                                      | 0                 | 0%               |
|                                                                                                               |                   |                  |
| Impact                                                                                                        | Number of answers | % over the total |
| Very high                                                                                                     | 1                 | 12.5%            |
| High                                                                                                          | 2                 | 25%              |
| Neutral                                                                                                       | 3                 | 37.5%            |
| Low                                                                                                           | 2                 | 25%              |
| Very low                                                                                                      | 0                 | 0%               |
|                                                                                                               |                   |                  |
| Averages                                                                                                      | Likelihood        | Impact           |
|                                                                                                               | 3                 | 3.25             |
|                                                                                                               |                   |                  |
| Risk index                                                                                                    | 9.75              |                  |

### 6.4.2 Rural “charging deserts”

#### Context and pathway to inequality

As the shift toward low-emission transport accelerates, access to reliable charging infrastructure becomes essential for both private and shared mobility. In rural and remote areas, however, the roll-out of charging stations is often slow, limited to main transit routes, or entirely absent, reflecting a market-driven investment logic that prioritises high-demand urban centres. At the same time, these same areas frequently suffer from inadequate public transport systems, whether due to infrequent service, limited coverage, lack of accessibility, or schedules poorly aligned with residents’ work and daily needs. This combination creates a dual constraint: residents who might consider switching to electric vehicles face practical barriers to doing so, while those without private cars cannot rely on public transport for

consistent, affordable mobility. Over time, these infrastructural gaps risk cementing a two-speed transition in mobility, where urban residents benefit from integrated, low-emission transport systems, and rural populations are left with limited and often more expensive options.

### Disadvantaged groups and negative impacts

The most affected by this hazard are rural residents, especially those on low incomes or without the flexibility to relocate to areas with better transport infrastructure. In places where public transport is inadequate, reliance on private vehicles is not a matter of choice but of necessity. For households unable to invest in electric vehicles due to high upfront costs and the absence of local charging infrastructure, this reliance means continued dependence on cars which carry higher running and maintenance costs and expose them to fuel price volatility.

These mobility constraints can also limit access to employment opportunities, education, healthcare, and other essential services, reinforcing cycles of social and economic disadvantage. The combination of infrastructure scarcity and weak public transport contributes to isolation and reduced quality of life, which can widen the gap between rural and urban living standards.

### Prioritization

| <b>Related Delphi Statement 1: Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption</b> |                          |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                         | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                          | 0                        | 0%                      |
| <b>High</b>                                                                                                                                               | 5                        | 55.6%                   |
| <b>Neutral</b>                                                                                                                                            | 3                        | 33.3%                   |
| <b>Low</b>                                                                                                                                                | 0                        | 0%                      |
| <b>Very low</b>                                                                                                                                           | 1                        | 11.1%                   |
|                                                                                                                                                           |                          |                         |
| <b>Impact</b>                                                                                                                                             | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                          | 3                        | 33.3%                   |
| <b>High</b>                                                                                                                                               | 3                        | 33.3%                   |
| <b>Neutral</b>                                                                                                                                            | 1                        | 11.1%                   |
| <b>Low</b>                                                                                                                                                | 2                        | 22.2%                   |
| <b>Very low</b>                                                                                                                                           | 0                        | 0%                      |
|                                                                                                                                                           |                          |                         |
| <b>Averages</b>                                                                                                                                           | <b>Likelihood</b>        | <b>Impact</b>           |
|                                                                                                                                                           | 3.33                     | 3.77                    |
|                                                                                                                                                           |                          |                         |
| <b>Risk index 1</b>                                                                                                                                       | 12.55                    |                         |

| <b>Related Delphi Statement 2: Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs</b> |                          |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                      | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                       | 3                        | 33.3%                   |
| <b>High</b>                                                                                                                                            | 3                        | 33.3%                   |
| <b>Neutral</b>                                                                                                                                         | 2                        | 22.2%                   |
| <b>Low</b>                                                                                                                                             | 1                        | 11.1%                   |
| <b>Very low</b>                                                                                                                                        | 0                        | 0%                      |
|                                                                                                                                                        |                          |                         |
| <b>Impact</b>                                                                                                                                          | <b>Number of answers</b> | <b>% over the total</b> |

|                     |                   |               |
|---------------------|-------------------|---------------|
| <b>Very high</b>    | 2                 | 22.2%         |
| <b>High</b>         | 3                 | 33.3%         |
| <b>Neutral</b>      | 3                 | 33.3%         |
| <b>Low</b>          | 1                 | 11.1%         |
| <b>Very low</b>     | 0                 | 0%            |
|                     |                   |               |
| <b>Averages</b>     | <b>Likelihood</b> | <b>Impact</b> |
|                     | 3.88              | 3.66          |
|                     |                   |               |
| <b>Risk index 2</b> | 14.2              |               |

**Total average risk index** = 13.375

### 6.4.3 Equity-blind implementation of Low-Emission Zones (LEZ) and restrictions on older vehicles

#### Context and pathway to inequality

The implementation of measures like Low-Emission Zones and other urban vehicle restrictions aim to reduce air pollution and greenhouse gas emissions by limiting or banning the use of older, more polluting vehicles in designated areas, typically city centres. While they are effective environmental tools, when introduced without compensatory policies they can disproportionately impact individuals and households who cannot afford to upgrade to compliant vehicles. The problem is increased when exemptions are limited and deadlines for compliance are short.

For those affected, the introduction of such zones can mean sudden restrictions on their ability to enter urban areas for work, education, healthcare, or family obligations. In the absence of viable public transport alternatives or financial support for vehicle replacement, these measures risk increasing spatial and social inequalities by privileging those who already have the means to adapt.

#### Disadvantaged groups and negative impacts

The most affected by this hazard include low-income drivers, and residents of peripheral and/or rural areas who rely on older vehicles to access urban services and opportunities. For low-income households, replacing a non-compliant vehicle with an electric or low-emission alternative is often financially impossible, and the lack of accessible subsidies schemes exacerbates this barrier. Those living in areas with poor public transport connections face additional challenges, as they cannot substitute private vehicle use with reliable and affordable alternatives. The negative impacts manifest in several ways: some are forced to take longer, more expensive journeys to avoid restricted zones; other risk losing access to jobs or essential services; and in the most extreme cases, individuals may withdraw from opportunities entirely, leading to reduced income and diminished quality of life due to social isolation. In this way, an environmental measure designed to promote cleaner air can inadvertently widen socio-economic divides.

#### Prioritization

| <b>Related Delphi Statement 1: Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption</b> |                          |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| <b>Likelihood</b>                                                                                                                                         | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>                                                                                                                                          | 0                        | 0%                      |
| <b>High</b>                                                                                                                                               | 5                        | 55.6%                   |
| <b>Neutral</b>                                                                                                                                            | 3                        | 33.3%                   |

|              |                   |                  |
|--------------|-------------------|------------------|
| Low          | 0                 | 0%               |
| Very low     | 1                 | 11.1%            |
|              |                   |                  |
| Impact       | Number of answers | % over the total |
| Very high    | 3                 | 33.3%            |
| High         | 3                 | 33.3%            |
| Neutral      | 1                 | 11.1%            |
| Low          | 2                 | 22.2%            |
| Very low     | 0                 | 0%               |
|              |                   |                  |
| Averages     | Likelihood        | Impact           |
|              | 3.33              | 3.77             |
|              |                   |                  |
| Risk index 1 | 12.55             |                  |

|                                                                                                                                                        |                   |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| <b>Related Delphi Statement 2: Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs</b> |                   |                  |
| Likelihood                                                                                                                                             | Number of answers | % over the total |
| Very high                                                                                                                                              | 3                 | 33.3%            |
| High                                                                                                                                                   | 3                 | 33.3%            |
| Neutral                                                                                                                                                | 2                 | 22.2%            |
| Low                                                                                                                                                    | 1                 | 11.1%            |
| Very low                                                                                                                                               | 0                 | 0%               |
|                                                                                                                                                        |                   |                  |
| Impact                                                                                                                                                 | Number of answers | % over the total |
| Very high                                                                                                                                              | 2                 | 22.2%            |
| High                                                                                                                                                   | 3                 | 33.3%            |
| Neutral                                                                                                                                                | 3                 | 33.3%            |
| Low                                                                                                                                                    | 1                 | 11.1%            |
| Very low                                                                                                                                               | 0                 | 0%               |
|                                                                                                                                                        |                   |                  |
| Averages                                                                                                                                               | Likelihood        | Impact           |
|                                                                                                                                                        | 3.88              | 3.66             |
|                                                                                                                                                        |                   |                  |
| Risk index 2                                                                                                                                           | 14.2              |                  |

**Total average risk index** = 13.375

#### 6.4.4 Accessibility gaps in digital mobility platforms and booking systems

##### Context and pathway to inequality

As transport services become increasingly integrated and digitalized, platforms for booking and paying for trips (like public transport or shared mobility) are often designed with assumptions about users' abilities and access to technology. When these systems lack consistent universal design features they create hidden barriers to participation. These gaps can prevent certain groups from accessing services entirely or force them into more expensive, less convenient travel options. The risk grows as more transport providers phase out in-person ticketing or cash payment systems, making digital interaction a prerequisite for using basic mobility services.



### Disadvantaged groups and negative impacts

The disadvantaged groups derived from this hazard are people with disabilities or sensory challenges (e.g. those with visual, auditory, cognitive, or motor impairments), and people with lower digital literacy. For individuals with visual impairments, poorly designed apps without screen reader functionality can make essential journey planning impossible. Those with reduced dexterity can find touchscreen-based interactions physically difficult. Older adults, even when willing to use digital tools, may have problems due to a combination of unfamiliarity with newer interfaces or fast-changing application updates.

These accessibility barriers translate into tangible negative impacts: exclusion from integrated and often more affordable mobility offers, dependence on others to arrange travel, and even in some cases inability to reach certain destinations if no non-digital options exist. Such exclusion can reduce independence and access to basic services or social activities, which leads to increasing existing inequalities in mobility and participation.

### Prioritization

| Related Delphi Statement 1: New business models such as Mobility-as-a-Service (MaaS) models will be a reality in 2050 |                   |                  |
|-----------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                            | Number of answers | % over the total |
| Very high                                                                                                             | 1                 | 11.1%            |
| High                                                                                                                  | 5                 | 55.5%            |
| Neutral                                                                                                               | 2                 | 22.2%            |
| Low                                                                                                                   | 1                 | 11.1%            |
| Very low                                                                                                              | 0                 | 0%               |
|                                                                                                                       |                   |                  |
| Impact                                                                                                                | Number of answers | % over the total |
| Very high                                                                                                             | 0                 | 0%               |
| High                                                                                                                  | 4                 | 44.4%            |
| Neutral                                                                                                               | 5                 | 55.5%            |
| Low                                                                                                                   | 0                 | 0%               |
| Very low                                                                                                              | 0                 | 0%               |
|                                                                                                                       |                   |                  |
| Averages                                                                                                              | Likelihood        | Impact           |
|                                                                                                                       | 3.66              | 3.44             |
|                                                                                                                       |                   |                  |
| Risk index 1                                                                                                          | 12.59             |                  |

| Related Delphi Statement 2: By 2050, subscription-based “car as a service” models will be the default, instead of private cars |                   |                  |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                     | Number of answers | % over the total |
| Very high                                                                                                                      | 0                 | 0%               |
| High                                                                                                                           | 2                 | 22.2%            |
| Neutral                                                                                                                        | 2                 | 22.2%            |
| Low                                                                                                                            | 4                 | 44.4%            |
| Very low                                                                                                                       | 1                 | 11.1%            |
|                                                                                                                                |                   |                  |
| Impact                                                                                                                         | Number of answers | % over the total |
| Very high                                                                                                                      | 0                 | 0%               |
| High                                                                                                                           | 3                 | 33.3%            |

|              |            |        |
|--------------|------------|--------|
| Neutral      | 5          | 55.5%  |
| Low          | 0          | 0%     |
| Very low     | 1          | 11.1%  |
|              |            |        |
| Averages     | Likelihood | Impact |
|              | 2.55       | 3.11   |
|              |            |        |
| Risk index 2 | 7.93       |        |

**Total average risk index** = 10.26

#### 6.4.5 Automation and restructuring in automotive/logistics with uneven reskilling

##### Context and pathway to inequality

The integration of robotics, artificial intelligence, digital twins, and other advanced technologies is transforming production lines and supply chain management, often resulting in the reduction or redefinition of certain job roles. While these changes are intended to improve efficiency and reduce emissions, they can displace workers whose skills are no longer aligned with emerging demands.

In the absence of comprehensive and accessible training programs that anticipate sectoral shifts, many employees, particularly those in low-skilled roles, are left with limited options for redeployment. Restructuring can also lead to wage compression, with some workers moving to lower-paid positions despite having years of sector-specific experience.

##### Disadvantaged groups and negative impacts

The disadvantaged groups most affected are older and low-skilled workers in the automotive and logistics sectors, particularly those employed in manufacturing and supply. Older workers face significant barriers to retraining (such as the challenge of adapting to digital tools and processes late in their careers or a reduced ability to relocate for new roles). Low-skilled workers, whose roles are most susceptible to automation, are often offered reskilling programmes that are generic or poorly aligned with actual labour market needs, resulting in a mismatch between training and available jobs. This can lead to negative impacts such as sudden job loss, long periods of unemployment, reduced benefits, less stability, or re-employment in positions with lower pay.

This not only undermines individual livelihood but also affects the economic health of communities heavily dependent on these industries, as local spending power declines and support services face greater demand. This hazard risks increasing structural unemployment and regional inequality, with technological innovation perceived as a driver of job insecurity.

##### Prioritization

| Related Delphi Statement: The automatization of the traditional automotive manufacturing jobs will impact blue-collar workers |                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Likelihood                                                                                                                    | Number of answers | % over the total |
| Very high                                                                                                                     | 5                 | 55.5%            |
| High                                                                                                                          | 2                 | 22.2%            |
| Neutral                                                                                                                       | 1                 | 11.1%            |
| Low                                                                                                                           | 1                 | 11.1%            |

|                     |                          |                         |
|---------------------|--------------------------|-------------------------|
| <b>Very low</b>     | 0                        | 0%                      |
|                     |                          |                         |
| <b>Impact</b>       | <b>Number of answers</b> | <b>% over the total</b> |
| <b>Very high</b>    | 3                        | 33.3%                   |
| <b>High</b>         | 3                        | 33.3%                   |
| <b>Neutral</b>      | 2                        | 22.2%                   |
| <b>Low</b>          | 1                        | 11.1%                   |
| <b>Very low</b>     | 0                        | 0%                      |
|                     |                          |                         |
| <b>Averages</b>     | <b>Likelihood</b>        | <b>Impact</b>           |
|                     | 4.22                     | 3.88                    |
|                     |                          |                         |
| <b>Risk index 2</b> | 16.37                    |                         |

## 7. Prioritized Hazards summary table

| Rank | Hazard                                                                                             | Sector                        | Disadvantaged Groups                                                         | Negative Impacts                                                                    | Risk index |
|------|----------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------|
| 1    | Increase of precariousness in seasonal/migrant agricultural work                                   | Agriculture and food          | Seasonal and migrant farm workers                                            | Exploitation, unsafe conditions, low wages                                          | 22.50      |
| 2    | Sustainability standards that small-scale farmers and producers are unable to comply with          | Agriculture and food          | Small-scale farmers with limited resources                                   | High compliance costs, exclusion from markets, consolidation of agricultural sector | 21.38      |
| 3    | Capital-intensive agri-tech that small producers can't finance                                     | Agriculture and food          | Small/medium farmers, young and older farmers with limited capital           | Declining competitiveness, farm closures, rural decline                             | 18.06      |
| 4    | Landlord-tenant split incentives in retrofits                                                      | Housing and built environment | Low / moderate-income tenants in old rentals                                 | Energy poverty, high bills, poor indoor conditions, locked in inefficient housing   | 17.20      |
| 4    | High adaptation costs for homeowners amid price-sensitive incentives                               | Housing and built environment | Low-income homeowners                                                        | Debt, incomplete retrofits, high bills, exposure to climate risks                   | 17.20      |
| 4    | Exclusion of migrants / refugees / undocumented people from housing subsidies or formal programmes | Housing and built environment | Migrants, refugees, undocumented people                                      | Poor housing, energy poverty, health risks, social exclusion                        | 17.20      |
| 7    | Automation and restructuring in automotive/logistics with uneven reskilling                        | Mobility and transport        | Older and low-skilled workers                                                | Job losses, unemployment, wage compression, insecurity                              | 16.37      |
| 8    | Digital-by-default energy management without adequate support                                      | Energy                        | Elderly, low digital literacy groups, low-income households without internet | Exclusion from cheaper tariffs, higher bills, dependency on intermediaries          | 15.63      |
| 9    | Digitalisation / automation of construction processes displacing low-skilled labour                | Housing and built environment | Older and low-skilled workers                                                | Job losses, unemployment, wage compression, insecurity                              | 15.56      |

|           |                                                                                            |                        |                                                                |                                                                                      |       |
|-----------|--------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------|-------|
| <b>10</b> | Rural “charging deserts” and weak public transport                                         | Mobility transport and | Rural residents, low-income households                         | Limited mobility, isolation, reduced access to services, higher costs                | 13.38 |
| <b>10</b> | Equity-blind implementation of Low-Emission Zones (LEZ) and restrictions on older vehicles | Mobility transport and | Low-income drivers, rural / peripheral residents               | Loss of access to jobs / services, increased inequality                              | 13.38 |
| <b>12</b> | Abrupt decarbonization closures in fossil-fuel/energy value chains                         | Energy                 | Older and low-skilled workers, dependent communities           | Job losses, unemployment, wage compression, insecurity                               | 11.61 |
| <b>13</b> | Accessibility gaps in digital mobility platforms and booking systems                       | Mobility transport and | People with disabilities, elderly, low digital literacy groups | Exclusion from services, dependence on others, reduced independence                  | 10.26 |
| <b>14</b> | High upfront cost of Electric Vehicles and complex/limited subsidies                       | Mobility transport and | Low-income households, credit-constrained individuals          | Reliance on polluting cars, high fuel costs, exclusion from low-emission zones       | 9.75  |
| <b>15</b> | Slow/Uneven grid upgrades and limited renewable infrastructure outside cities              | Energy                 | Rural/remote residents, low-income households                  | Higher energy costs, exclusion from renewable markets, reinforced spatial inequality | 9     |

## 8. Limitations

The analysis presented in this deliverable is subject to several limitations that must be acknowledged in order to contextualize the findings and guide the interpretation of results.

First, not all methodological elements initially foreseen under T3.3 were implemented as planned. In particular, the hackathon activity, originally conceived as a joint initiative with WP7, could not be carried out within the reporting period.

This was primarily due to a sequence of approved delays in the project schedule, which affected the alignment between WP3 and WP7. The hackathon was designed to take place after the Delphi survey and in close connection with WP7 dissemination activities, so that its results could be validated and further disseminated. However, the cumulative delays in the completion of preceding tasks (notably the Delphi rounds and scenario building) compressed the timeline, leaving insufficient time to properly prepare, organise and execute the hackathon before the deadline for this deliverable.

In addition, the scheduling of WP7 activities was postponed to the following reporting period, further preventing the joint organisation initially envisaged. The absence of this exercise limited the opportunities for real-time co-creation and participatory validation of the hazards identified, although the findings remain robust thanks to triangulation across interviews, literature review, and expert surveys.

Second, while the DoA contemplated the application of a small set of eight semi-structured interviews with Disadvantaged Groups in each of the four sectors of analysis during this task, the actual implementation differed. Instead, the research team conducted a total of 34 semi-structured interviews coordinated by UCM. Although this larger sample size enhanced the diversity of the perspectives gathered, it also created challenges for direct comparability with the originally designed methodology.

Third, the inherent uncertainty of the scenarios developed under T3.2 must be highlighted. Future-oriented analysis by definition involves a degree of speculation and simplification. The identification of hazards and Disadvantaged Groups is therefore directly linked to assumptions included in the scenarios and in expert judgment. As niches and socio-technical regimes evolve, the specific risks and vulnerabilities may change, potentially altering the hazards identified.

Finally, although the Delphi survey provided a structured mechanism to rank hazards based on likelihood and impact, the number of expert responses was limited. This constraint reduces the statistical robustness of the prioritization exercise and increases the potential influence of individual opinions on the aggregated results. Consequently, the risk index values should be interpreted as indicative trends rather than definitive rankings.

## 9. Conclusions

This deliverable has examined the hazards and negative impacts associated with the twin transition across the energy, mobility and transport, housing and built environment, and agriculture and food sectors. The analysis has provided a comprehensive detail of how different vulnerabilities translated into tangible risks for specific Disadvantaged Groups. The results highlight that hazards rarely occur in isolation; rather, they tend to interact and reinforce one another, generating cumulative disadvantage.

The findings demonstrate that the groups most exposed to hazards are often those already facing structural inequalities, such as low-income households, older workers, migrants and refugees, people with disabilities, and rural residents. In many cases, the negative impacts identified contribute to cycles of vulnerability that deepen social and economic divides. Importantly, the prioritization exercise revealed that hazards linked to agriculture and housing stand out as particularly pressing.

The results also show that digitalization, while often framed as a driver of empowerment and efficiency, can itself become a source of exclusion when implemented without adequate support. Similarly, environmental policies (such as decarbonization measures or housing retrofits) can inadvertently exacerbate inequalities if equity dimensions are not systematically integrated into their design.

A key conclusion is that the success of the twin transition is closely related not only to technological and environmental outcomes but also on social justice and fairness. As expressed before, the transition will be “fair or it will not be”. There is a real risk of eroding public trust and provoking resistance to the transition process if proactive policies to mitigate negative impacts are not implemented.

The insights from this deliverable, alongside D3.1 and D3.2, provide a critical basis for subsequent work in WP4. The evidence presented here will inform the design of policy measures and participatory processes aimed at ensuring that vulnerable and Disadvantaged Groups are not left behind.

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